

The Power of Paranoia
Privacy, Publicity, Causal Explosion, and the Institutions of Disciplined Distrust

Flyxion
Independent Researcher

Abstract

Privacy and publicity are commonly treated as opposing values: privacy conceals, whereas publicity exposes. This essay argues that both are responses to the same underlying condition—the severe difficulty of reliable judgment in causally dense environments. Public recording can preserve enormous quantities of evidence without producing timely understanding; indeed, the accumulation of video, testimony, metadata, and commentary may enlarge the interpretive search space faster than investigators can reduce it. Salient facts may take years or decades to move through hierarchies of attention and acquire legal significance. Under these conditions, suspicion is neither simply irrational nor sufficient for action. Its legitimate function is to keep open the possibility that appearances are incomplete. Legal requirements that appear cumbersome—warrants, evidentiary standards, adversarial testing, burdens of proof, and appellate delay—constitute an institutional form of disciplined paranoia. They assume that observers, officials, crowds, and technical systems may be mistaken or untrustworthy. By slowing the passage from suspicion to coercion, these constraints do not merely limit power; they create the protected uncertainty within which freedom becomes possible.

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Part I

Volume I: The Epistemology of Suspicion —
Part I. The Power of Paranoia

Chapter 1

Suspicion as an Epistemic Operator

Suspicion must first be separated from the other postures with which it is routinely confused: clinical paranoia, political paranoia, methodological skepticism, adversarial reasoning, and institutional distrust. Before any formalism, the book fixes three states that readers otherwise conflate under a single word:

naivety : the presented account is presumed sufficient,
disciplined paranoia : unrepresented causes remain possible,
pathological paranoia : every observation confirms hostile agency.

Disciplined paranoia is the book's subject, and it is subject to three constraints that separate it from its pathological neighbor: it must specify what evidence would weaken its suspicions, it must assign unequal plausibility to competing alternatives rather than treating every unexcluded history as equally live, and it must forbid converting mere possibility into punitive action. The remainder of the chapter gives this trichotomy its operator-level content. The central object of the chapter is the epistemic operator

$$\Pi : \mathcal{E} \longrightarrow 2^{\mathcal{H}},$$

where $\mathcal{E}$ is the space of available evidence and $\mathcal{H}$ is the space of causally possible histories. Rather than forcing a single explanation from a partial record, $\Pi(E)$ preserves the set of histories not yet excluded by E .

The danger is premature closure: any selection map

$$C_\tau : \Pi(E_\tau) \longrightarrow \hat{H}_\tau$$

that chooses one history before the available evidence has passed through the relevant verification procedures. In these terms, disciplined paranoia enlarges or preserves $\Pi(E)$ subject to the three constraints above; pathological paranoia refuses exclusion altogether and treats every member of $\Pi(E)$ as equally actual or equally accusatory, which is exactly the failure of the second constraint (unequal plausibility) and the third (no license to act on mere possibility).

The governing distinction is therefore not between trust and doubt, but between epistemic stages:

suspicion $\neq$ belief $\neq$ proof $\neq$ authorization.

That distinction is sharpened by the first non-implication chain of the book:

$$\text{Possible}(H) \not\Rightarrow \text{Probable}(H) \not\Rightarrow \text{Proven}(H) \not\Rightarrow \text{Actionable}(H).$$

Suspicion is thus the discipline of keeping possibility visible without mistaking possibility for either proof or permission.

Chapter 2

Why Appearances Underdetermine Causes

Appearances underdetermine causes because what is observed is a projection of a richer latent process:

$$Y_t = \mathcal{O}(X_t, \eta_t),$$

where X_t is the causally relevant state, $\mathcal{O}$ is an observation channel, and η_t collects noise, occlusion, selection, framing, compression, and measurement error. What arrives as evidence is therefore already filtered by a medium and a circumstance of capture.

The inverse problem is generally non-identifiable:

$$\mathcal{O}(X_1, \eta_1) = \mathcal{O}(X_2, \eta_2) \quad \text{while} \quad X_1 \neq X_2.$$

An authentic video, statement, or measurement can therefore be real without being causally decisive. Truncated recordings, ambiguous agency, hidden antecedents, misleading timestamps, and observational equivalence all show how distinct histories can survive the same surface appearance.

This chapter accordingly distinguishes falsified evidence from incomplete evidence, decontextualized evidence, and causally insufficient evidence. Public argument often collapses those four categories into the blunt opposition between the “real” and the “fake,” but the epistemic problem is broader: a datum may be genuine and still fail to determine what happened.

Chapter 3

The First Available Story

If interpretations arrive sequentially as $H_1, H_2, \dots, H_n$, the first available story acquires advantages before truth has been settled. Early narratives benefit from anchoring, repetition, indexing, search ranking, organizational memory, and resource allocation. Let $w_i(t)$ be the institutional weight assigned to hypothesis H_i . A simple reinforcement process is

$$w_i(t+1) = (1 - \lambda)w_i(t) + \lambda s_i(t) + \rho a_i(t),$$

where $s_i(t)$ is evidentiary support and $a_i(t)$ is accumulated attention. When ρ is large, attention becomes partially self-validating: uptake helps create the very standing later interpreted as confirmation.

The chapter names this persistence narrative hysteresis. Even when the underlying evidence changes, public judgment can remain trapped in an earlier interpretive basin because the first story has already reorganized memory, testimony, and institutional effort around itself. The task of disciplined suspicion here is not to deny narrative need, but to resist treating temporal precedence as epistemic entitlement.

Chapter 4

Paranoia, Skepticism, and Conspiracy

A conspiracy hypothesis is not disqualified merely because it proposes coordination; many institutions are coordinated by definition. The decisive question is instead whether the hypothesis remains open to exclusion. Suspicion becomes epistemically pathological when every possible observation is absorbed as confirmation:

$$P(E \mid H) \approx P(E' \mid H)$$

for mutually incompatible observations E and E' . At that point the hypothesis no longer risks contact with the world, because nothing that happens can count against it.

The chapter therefore distinguishes open suspicion from self-sealing suspicion. Open suspicion specifies possible falsifiers, competing explanations, and thresholds of action. Self-sealing suspicion treats missing evidence as proof of concealment and counterevidence as proof of infiltration. The issue is not whether hidden coordination is imaginable, but whether inquiry preserves a meaningful relation between what is observed and what one is prepared to conclude.

Chapter 5

Distrust Without Accusation

Distrust need not begin in accusation. Institutions can be designed around fallibility while remaining agnostic about which failure, if any, has occurred. A protocol may assume

$$P(\text{error}) > 0, \quad P(\text{capture}) > 0, \quad P(\text{misrepresentation}) > 0$$

without claiming in advance that any particular actor is malicious. The point is to structure action under unresolved uncertainty rather than to smuggle in a hidden verdict of guilt.

On that basis, institutional distrust is expressed through redundancy, separation of powers, audit, cross-examination, reproducibility, conflict-of-interest rules, and appeal. These mechanisms encode a general proposition: no observer, witness, office, or archive should be treated as a transparent channel to reality. Distrust is therefore not identical with condemnation. It is a way of distributing vulnerability when the evidence does not yet justify accusation but does justify procedural safeguards.

Part II

Volume I: The Epistemology of Suspicion — Part II. Events, Records, and Reconstructions

Chapter 6

The Event Is Not the Record

The event is not the record because the ontology of occurrence and preservation already contains several transformations:

$$X \xrightarrow{\tau} T \xrightarrow{\kappa} R \xrightarrow{\mu} M.$$

Here X is the event, T its trace, R a deliberately or automatically retained record, and M a representation that has been interpreted, edited, indexed, or narrated. Each arrow can lose information, add structure, or alter what later counts as salient.

The distinctions matter. A trace is a physical consequence of an event; a record is a trace that survives in some preserved form; a representation is a record already shaped for reading, retrieval, or judgment. Preservation therefore does not entail accessibility:

$$\text{Recorded}(x) \not\Rightarrow \text{Retrievable}(x) \not\Rightarrow \text{Recognized}(x).$$

A fact can be recorded yet buried, retrievable yet unnoticed, and noticed yet misrecognized. This chapter's preface to the rest of Part II is that documentary survival must never be mistaken for transparent presence.

Chapter 7

Provenance and the Chain of Transformation

Once the event is separated from its record, provenance must be modeled as a transformation history rather than a badge of origin. A record's provenance is represented as a directed acyclic graph whose nodes are transformations and whose edges carry custody relations. Confidence therefore attaches not simply to a final artifact, but to the integrity of the path by which it was observed, stored, transmitted, edited, and retrieved.

For a path $p = (e_1, \dots, e_k)$, one provisional integrity measure is

$$I(p) = \prod_{j=1}^k r(e_j),$$

where $r(e_j) \in [0, 1]$ measures the reliability of transformation e_j . The chapter uses that formalism only as a starting point, since multiplicative trust can conceal correlated failure and common-source dependence. The real lesson is that evidential confidence must be distributed across the chain of transformation, not projected backward from the last visible artifact.

Chapter 8

Recording as Selection

No recording system captures everything. A camera selects a field of view, frame rate, spectral range, resolution, exposure, compression scheme, storage duration, and metadata policy; a form selects prompts, fields, and categories. The recorded object is therefore not simply the event itself but

$$R = S_{\theta}(X),$$

where θ is a selection regime that determines what becomes recordable at all.

This chapter introduces selection geometry: the pattern of invisibilities produced when a recording apparatus is optimized for one class of events rather than another. Surveillance designed to identify faces may fail to preserve gait, peripheral interaction, audio context, or environmental causation. The important question is thus not only whether a record was altered after capture, but what the architecture of capture rendered visible or invisible from the start.

Chapter 9

Compression, Salience, and Loss

Records rarely remain in their initial form. They move through successive compressions,

$$R_0 \rightarrow R_1 \rightarrow \cdots \rightarrow R_k,$$

by which raw events become footage, footage becomes selected clips, clips become reports, reports become headlines, and headlines become remembered claims. Each stage reduces bandwidth while increasing apparent intelligibility, making coordination easier at the cost of background detail and alternative lines of inquiry.

To track that cost, the chapter defines a distortion functional

$$D(R_i, R_{i+1}; Q),$$

where Q is the class of questions later asked of the record. Compression that is harmless for one question may destroy the evidence needed for another. No summary is universally sufficient, because every gain in salience is also a wager about which future questions will matter.

Chapter 10

Reconstruction as an Inverse Problem

Historical and legal reconstruction are inverse problems under partial observation. Given evidence E , one seeks

$$\hat{H} = \arg \max_{H \in \mathcal{H}} P(H \mid E).$$

But that posterior never arises from evidence alone. It depends on the hypothesis space $\mathcal{H}$, the prior distribution over candidate histories, the likelihood model connecting traces to causes, and the exclusion or omission of unknown alternatives.

Bayesian language therefore does not eliminate judgment; it relocates judgment into model construction. Two investigators can agree on the surviving record and still disagree about priors, admissible hypotheses, or which absences count as informative. The chapter closes Part II by treating reconstruction as disciplined inference under irreversible loss: rigor comes not from pretending the inverse problem disappears, but from making the structure of the inference explicit.

Part III

Volume II: Causal Explosion — Part III. The Combinatorics of Context

Chapter 11

Causal Graphs and Expanding Neighborhoods

This chapter begins from the claim that every event is embedded in a causal graph $G = (V, E)$. To understand a node v , inquiry expands its backward causal neighborhood

$$B_k(v) = \{u \in V : d(u, v) \leq k\}.$$

[Definition] Here $d(u, v)$ is graph distance and k the inquiry radius. If the average branching factor is b , then the size of that neighborhood grows as

$$|B_k(v)| \approx \sum_{j=0}^k b^j = \frac{b^{k+1} - 1}{b - 1}.$$

[Toy model] This average-branching approximation makes causal explosion visible before case-specific detail is added. Even modest values of b and k therefore generate an enormous candidate context; this is the elementary mathematics of causal explosion.

The practical question is not whether an investigator can map the whole graph, but how far the relevant neighborhood must expand before attribution is warranted. A visible trigger rarely exhausts the explanation, because background conditions, mediating institutions, incentives, and omissions can all lie inside a larger $B_k(v)$ and redirect responsibility once they are brought into view. The chapter treats disciplined suspicion as a rule for enlarging adjacency until further expansion stops materially altering the causal account.

Chapter 12

Combinatorial Histories

Once causal neighborhoods expand, histories multiply rapidly. If n potentially relevant events can be ordered in multiple ways, the number of candidate temporal structures may approach

$$n!.$$

[Toy model] This factorial count is a stylized lower bound on historical multiplicity. If each of m variables has q plausible values, the joint assignment space contains

$$q^m$$

[Toy model] This product-space count isolates how hidden-state variation outruns any single visible outcome. possibilities; if directed causal edges are uncertain, the number of admissible graphs grows super-exponentially. This is the combinatorial setting in which legal and political reconstruction actually operates.

The point is not that investigators literally enumerate every candidate history. The point is that every workable inquiry must impose pruning rules on an underlying space that is vastly larger than the official narrative usually admits. Those pruning rules are epistemically decisive, because they determine which truths remain discoverable and which pathways are discarded before they can even be tested.

Chapter 13

Counterfactual Branching

Responsibility depends on counterfactual branching: to say that an act mattered is to show that nearby interventions change the outcome, so that

$$Y_{do(A=a)} \neq Y_{do(A=a')}.$$

[Definition] This intervention contrast fixes the book’s difference-making criterion for saying that an act mattered. Legal events, however, rarely involve a single isolated switch. They involve multiple agents, delayed consequences, feedback, strategic adaptation, and the familiar problems of structural causal models, potential outcomes, actual causation, overdetermination, preemption, and omissions.

To keep this branching disciplined, the chapter introduces the counterfactual branching functional

$$\mathcal{B}(X) = \sum_{a \in \mathcal{A}} |\Omega_{do(a)}|,$$

[Definition] This branching functional names the disciplined comparison class for nearby alternatives. which counts the materially distinct outcome spaces generated by plausible interventions. The aim is not to indulge fantasy about unreal worlds, but to compare the actual path with nearby mechanism-grounded alternatives. Negligence, intent, prevention, and responsibility all depend on which branches are live, how many of them there are, and which changes would have made a difference.

Chapter 14

Ecological Judgment

Ecological judgment begins from the fact that context can change the causal type of an act. What looks local may depend on technical systems, household arrangements, institutional incentives, medicine, psychology, labor, law, or environmental conditions. To formalize that embedding, the chapter introduces a multiscale model

$$X^{(0)} \subset X^{(1)} \subset \dots \subset X^{(L)},$$

[Definition] This multiscale embedding fixes the chapter's vocabulary for ecological levels of context. where level 0 is the immediately visible action and higher levels represent interpersonal, institutional, infrastructural, historical, and ecological surroundings.

The central problem is to choose the smallest scale at which the event becomes intelligible without letting inquiry expand indefinitely. A rule, heuristic, burden, or presumption should therefore be judged by how well it fits the environment described at the relevant level of $X^{(L)}$, not by an abstract fantasy of omniscience. Paranoia is rational here only when it is ecologically calibrated: enough context to identify the act correctly, not so much that explanation dissolves into boundless scenery.

Chapter 15

Computational Irreducibility and Legal Fact

Some legally relevant facts are governed by computational irreducibility: their consequences cannot be inferred substantially faster than the underlying process can unfold. In such domains, long-term perspective cannot be replaced by a sufficiently clever snapshot, because the legally important structure appears only through staged interaction rather than through an *ex ante* summary.

The chapter separates strong claims of undecidability from the more practical limits of intractability. It focuses on NP-hard search, model uncertainty, chaotic sensitivity, strategic opacity, and the impossibility of specifying a complete state description in advance. The lesson is not mystification about complexity, but procedural humility: when social processes can only be learned by running them, law must rely on revisable findings, staged observation, and institutions that can absorb information that honest inquiry could not have front-loaded.

Chapter 16

Unknown Unknowns

Ordinary probability models distribute likelihood over a given hypothesis space, but inquiry also faces unknown unknowns: cases in which the relevant explanation is absent from the model itself,

$$H^* \notin \mathcal{H}.$$

[Definition] This condition defines model incompleteness rather than ordinary uncertainty over listed options. This chapter distinguishes risk, uncertainty, ambiguity, ignorance, and model incompleteness. Its claim is that many investigative failures arise not from misweighting listed options, but from treating the current menu of options as complete when it is not.

That is why legal procedure cannot be satisfied with a single closed pass through a case. If $\mathcal{H}$ may omit the truth, then reopening, appeal, anomaly reporting, and adversarial challenge are not luxuries but safeguards against model closure. The right lesson is disciplined openness to ontology revision, not limitless suspicion: the task is to preserve mechanisms by which an unrepresented hypothesis can become thinkable and then testable.

Part IV

Volume II: Causal Explosion — Part IV. Time, Delay, and the Maturation of Evidence

Chapter 17

Evidentiary Time

This chapter defines evidentiary time by separating event time t_e , recording time t_r , and recognition time t_s . A detail may be captured immediately and yet matter only much later:

$$t_e \leq t_r \ll t_s.$$

[Definition] This temporal ordering fixes the chapter’s vocabulary for evidentiary latency. That gap is the latency of evidence. It is not an accidental inconvenience but a structural variable in inquiry, because what can responsibly be believed depends on where one stands relative to those three times.

Not every source of delay bears on that gap in the same way, so the Part fixes four kinds before treating any of them as a virtue. Procedural delay lets evidence mature between t_r and t_s , and is the only kind inherently connected to epistemic restraint. Capacity delay arises because an institution lacks the investigators, instruments, or expertise to process what it already holds. Strategic delay is imposed by an actor who benefits when attention, witnesses, or resources dissipate before t_s . Pathological delay results from institutional indifference or dysfunction with no compensating evidentiary work occurring in the interval. Whenever a later chapter credits delay with protecting judgment, the claim being made is about procedural delay specifically; the other three can destroy the very freedom this Part is trying to explain, and each should be named rather than left to hide inside a generic appeal to patience.

Evidence therefore has a temporal ecology. Traces appear, decay, corroborate, and change significance on different schedules, so a claim that is weakly warranted early can become strongly warranted later without the underlying event changing at all. The chapter treats delay as both a cost and a condition of truth, and asks how institutions should act when judgment must often be made before recognition time has caught up with recording time.

Chapter 18

Millions of Hours

Modern inquiries often confront millions of hours of record rather than a scarcity of traces. If a public event produces V hours of video and reviewers can process footage at an effective rate r , then naïve serial review requires

$$V/r$$

[Toy model] This serial-review quotient isolates the throughput bottleneck before coordination costs are added. hours of attention. Parallel review lowers wall-clock time, but only by introducing coordination burdens, duplication, inconsistent labeling, and escalation costs. This is capacity delay in its purest form: the institution withholds no evidence and gains nothing from waiting, it simply does not yet have enough throughput to convert what it holds into judgment.

The chapter therefore models evidentiary review as a queue. Evidence items arrive at rate λ , reviewers process them at rate μ , and only a fraction p are escalated for higher scrutiny. When arrival outstrips effective review capacity, backlog grows even under comprehensive recording. The lesson is that duration can be probative while still being operationally hard to harvest: a society may capture immense evidence and yet fail to convert it into timely judgment.

Chapter 19

Hierarchies of Saliency

A crucial detail often must survive a hierarchy before it can matter in judgment:

$$E_0 \rightarrow E_1 \rightarrow \cdots \rightarrow E_L.$$

[Definition] This chain defines the chapter's levels of evidentiary transmission. If at each level i it survives with probability p_i , then the probability that it reaches the final decision-maker is

$$P_{\text{reach}} = \prod_{i=0}^{L-1} p_i.$$

[Toy model] This reach probability is a stylized filter model rather than a fitted measurement law. This is the formal core of hierarchies of saliency: visibility is filtered step by step rather than given all at once.

The consequence is that important evidence may remain practically invisible even when it is public. Filters, triage, editorial selection, bureaucratic referral, algorithmic ranking, and expert validation all alter the p_i 's, sometimes elevating spectacle and sometimes burying what is most probative. The chapter studies how institutions can redesign those filters so that saliency becomes a tool of disciplined suspicion rather than a conveyor belt for distortion.

Chapter 20

Discovery Curves

This chapter models discovery curves by letting $F(t)$ denote the fraction of ultimately relevant evidence recognized by time t . Different inquiries produce different shapes: exponential, logistic, power-law, and punctuated curves all describe ways in which recognition can accumulate. Sometimes a late archival find or forensic breakthrough creates a discontinuity,

$$F(t^*+) - F(t^*-) \gg 0,$$

[Toy model] This jump condition sketches a late-breakthrough discovery curve without claiming a universal fit. showing that evidentiary maturation is not always smooth.

The institutional problem is how to behave when $F(t)$ remains far from saturation. Slow confirmation is not always weakness, and rapid early uptake is not always vindication. Resource allocation, investigative persistence, and reopening rules should be tuned to the expected shape of the curve in the relevant domain, because the tempo of learning is itself part of the evidence.

Chapter 21

The Half-Life of an Initial Narrative

Initial stories persist because memory, institutions, and reputation keep feeding them after their first presentation. The chapter models this half-life of an initial narrative by

$$N(t) = N_0 e^{-\delta t} + \int_0^t \rho(s) e^{-\delta(t-s)} ds,$$

[Toy model] This reinforcement-decay recursion illustrates narrative persistence without pretending to measure any one real system. where $\rho(s)$ represents reinforcement through repetition. Narrative authority therefore decays only partially on its own; it is continually replenished by social and institutional investment.

Correction is not symmetrical with accusation. Later evidence enters an attention environment already structured by the original narrative, so even good counterevidence must fight the installed machinery of paperwork, prestige, memory, and moral convenience. Some of that resistance is reinforced deliberately: an actor benefited by the original story can widen $\rho(s)$ or otherwise stretch the interval before t_s , which is strategic delay rather than the procedural delay this Part otherwise defends, and it should be named as such whenever it appears. The chapter asks how procedures can shorten the narrative's half-life without destroying coordination, so that revisability remains real rather than merely ceremonial.

Chapter 22

Statutes, Reopening, and Finality

Law must reconcile delayed learning with the need for closure. This chapter models that problem through the loss function

$$L(T) = L_{\text{premature}}(T) + L_{\text{delay}}(T),$$

[Toy model] This loss decomposition is a stylized design model for closure rather than a calibrated doctrinal metric. where T is the time of closure. Finality reached too early locks in avoidable error; finality deferred indefinitely destroys the security and administrability that judgment requires.

The optimal T is therefore neither immediate nor infinite. It varies with the reversibility and severity of the action under review, and it explains why statutes, reopening standards, clemency, and other second-look procedures should be understood as design variables rather than mere doctrinal leftovers. A closure rule set by $L(T)$ is distinct from a closure that never arrives because no one revisits the file: the latter is pathological delay, produced by indifference or dysfunction rather than by any evidentiary work being done in the interval, and it is not an instance of the restraint this chapter otherwise defends. The chapter ends the Part by treating finality as a problem of structured corrigibility: how to remain open to matured evidence without making decision impossible.

Part V

Volume III: Attention and Publicity — Part
V. Publicity Without Comprehension

Chapter 23

Visibility Is Not Knowledge

The chapter begins by rejecting the identification

more recording = more knowledge.

A more adequate relation is

$$K = f(V, I, C, A),$$

[Toy model] where V is evidentiary volume, I indexing quality, C contextual integrity, and A available attention. The point is not that visibility is worthless, but that knowledge depends on the interaction of these variables. If V rises while I , C , and A are held fixed, the probability of locating and correctly interpreting a relevant item can fall rather than rise.

Visibility multiplies appearances without by itself settling cause, relevance, or evidentiary weight. A regime of disclosure can therefore leave observers overwhelmed, manipulable, or falsely confident. The volume opens by marking the distinction between what can be seen and what can be responsibly known, and by motivating the mediating work later chapters assign to indexing, attention-allocation, contextual boundaries, and organized collective inquiry.

Chapter 24

The Evidentiary Haystack

Suppose N records contain only k relevant items. Under random inspection, expected search cost scales like N/k . Mere disclosure can therefore bury evidence under mass even before anyone argues about what it means. Salience tools do not dissolve the problem, because the features used to rank evidence already encode a theory of relevance; they help only by presuming, and thereby partly fixing in advance, what counts as worth finding.

This yields the paradox of search: to find the evidence that changes the theory, one often needs a theory capable of recognizing the evidence. The chapter treats the haystack not as a natural fact about abundance but as an institutional design problem. Duties to disclose are incomplete unless they are joined to preservation, indexing, provenance, and forms of organization that make unexpected evidence retrievable rather than merely available.

Chapter 25

Attention as a Scarce Public Resource

Public attention is modeled as a finite attention allocation vector

$$\mathbf{a} = (a_1, \dots, a_n), \quad \sum_{i=1}^n a_i \leq A.$$

[Definition] Publicity does not abolish scarcity; it organizes competition for A . Because every increase in salience somewhere spends capacity unavailable elsewhere, agenda-setting, repetition, and timing become epistemic mechanisms rather than merely rhetorical ones.

Powerful actors may therefore dominate not only by suppressing information, but by capturing or diluting attention shares within the common budget A . Equal formal access to speak does not equal equal access to uptake. The chapter reframes attention as public infrastructure and motivates triage, watchdog specialization, and anti-monopolization norms for the collective sensorium.

Chapter 26

Flooding, Noise, and Plausible Distraction

Censorship and flooding are dual strategies: censorship decreases accessible evidence, while flooding decreases the effective signal-to-attention ratio. If relevant signal has mass S , distracting material has mass N , and attention is bounded, effective discoverability may scale as

$$D \propto \frac{S}{S + N}.$$

[Toy model] The point is structural rather than psychological: when noise grows, it consumes the same interpretive budget that serious inquiry requires.

The chapter studies spam, document dumps, duplicated footage, performative transparency, adversarial disclosure, and archives that are technically accessible but practically unusable. Flooding preserves deniability because nothing need be hidden; what matters is simply prevented from remaining coherent long enough to bite.

Chapter 27

Algorithmic Saliency

Ranking systems implement a saliency function

$$\sigma(x \mid u, t, \theta),$$

[Definition] conditioned on user u , time t , and model parameters θ . What rises through the hierarchy is therefore not what is intrinsically important, but what is important relative to an objective function. Recurrence can be an artifact of optimization rather than evidentiary force.

The chapter distinguishes engagement saliency, investigative saliency, legal saliency, scientific saliency, and historical saliency. Their divergence explains why publicly dominant evidence need not be probatively strong. Any politics of publicity must therefore scrutinize the objective functions and feedback loops that allocate notice before explicit judgment begins.

Chapter 28

Collective Investigation

Large crowds can parallelize discovery, but they also generate shared priors, copied mistakes, harassment, false identification, and narrative cascades. The chapter models collective investigation as a distributed search process with non-independent errors. Its effective number of independent investigators is not the head count n , but approximately

$$n_{\text{eff}} = \frac{n}{1 + (n - 1)\rho},$$

[Toy model] where ρ is average error correlation. As ρ rises, crowds become less like many checks and more like one amplified misreading.

The argument therefore favors organized collective inquiry rather than romantic crowd omniscience. Reliable public knowledge requires institutions and norms that lower ρ : role differentiation, adversarial checking, preserved records, and explicit thresholds for moving from clue to claim. Publicity becomes epistemically useful only when scattered observers can be coordinated without being collapsed into a single cascade. Those same thresholds become unmeetable burdens when they are used to discount dispersed but convergent clues until only already-authorized investigators can move from suspicion to claim.

Part VI

Volume III: Attention and Publicity — Part VI. Privacy as Boundary and Capacity

Chapter 29

Privacy Beyond Secrecy

Privacy is defined here not as concealment but as control over the conditions under which information about a person crosses contextual boundaries. Those conditions govern collection, inference, combination, retention, retrieval, interpretation, and action. The chapter distinguishes informational privacy, decisional privacy, spatial privacy, relational privacy, inferential privacy, and temporal privacy, and treats each as a mode of boundary maintenance rather than a mere reservoir of hidden facts.

This broader account explains why the slogan “nothing to hide” misses the point. Privacy is a capacity for agency, rehearsal, role differentiation, and selective disclosure under non-totalizing conditions. It is therefore compatible with, and often necessary for, responsible inquiry: a society can demand accountability from power without demanding exhaustive legibility from persons.

Chapter 30

The Mosaic Problem

The mosaic problem begins from the fact that individually harmless disclosures need not remain harmless in aggregate. If disclosures $D_1, \dots, D_n$ interact, aggregate privacy risk is generally nonadditive:

$$R(D_1 \cup \dots \cup D_n) \neq \sum_i R(D_i).$$

More fully,

$$R(D_{1:n}) = \sum_i R(D_i) + \sum_{i < j} I_{ij} + \sum_{i < j < k} I_{ijk} + \dots,$$

[Toy model] where the synergistic inference terms $I_{ij}, I_{ijk}, \dots$ register what only combinations reveal.

The chapter uses this expansion to connect privacy loss to combinatorial causal inference. Item-by-item sensitivity tests miss the level at which harm emerges, because no single datum need contain the invasion later produced by linkage, accumulation, and recombination. Privacy can be lost not only by revelation, but by composition. The same resistance to recombination becomes concealment when aggregation rules are drawn so broadly that patterns of official or corporate power cannot be assembled either.

Chapter 31

Contextual Integrity

Contextual integrity represents an information flow as

$$\tau = (s, r, \alpha, c, p),$$

[Definition] where s is sender, r recipient, α information type, c context, and p transmission principle. On this account, many privacy violations are not disclosures of secrets at all, but unauthorized changes in τ : the same content crosses into a different role-structure, purpose, or principle of transmission.

This framework replaces the crude public/private binary with a more exact question: who may know what, in which role, in which context, and under which transmission constraints? Information appropriate in one setting may become harmful when moved into another, even when nothing about the underlying fact changes. Privacy protects intelligibility by preserving these context-sensitive norms of flow. The same norm-preserving restrictions become concealment when institutions invoke context to block adversarial scrutiny of actors exercising coercive or allocative power.

Chapter 32

Privacy as Protection Against Model Capture

Model capture names the condition in which an individual’s actionable institutional representation becomes difficult to inspect, correct, or escape. Formally, the danger appears when

$$M_t(x) \rightarrow A_t(x)$$

[Definition] when there is no adequate route from the represented person back to M_t . Publicly available traces can then be pooled into a durable model that shapes access to employment, insurance, credit, mobility, reputation, or legal scrutiny before the subject can contest what pattern is being used.

The central harm is not observation alone, but the externalization and operationalization of a person-pattern as someone else’s asset. Once cross-context traces are stabilized into a predictive representation, self-presentation becomes reactive and suspicion becomes effectively unilateral. Privacy therefore functions as protection against the capture, refinement, and deployment of these models. The same barriers to capture become concealment when institutions claim proprietary or security privacy for high-stakes models while still acting on them against the people they score.

Chapter 33

The Right to Remain Underdescribed

The chapter introduces underdescription as a political good. Human freedom partly depends on not being perfectly resolved into a predictive object, and a person should not be forced to exhaustively explain every anomaly merely because extensive records make such explanation technically demandable. The relevant right is not a right to deceive; it is a limit on the presumption that whatever can be inferred may therefore be legitimately used.

Placed after model capture, the point is sharper: compulsory completeness turns persons into administratively convenient dossiers. A right to remain underdescribed preserves revisability, experimentation, and context-specific self-presentation by refusing the demand that every trace be made permanently commensurable with every institutional purpose. The same protection becomes concealment when officials or powerful organizations claim a comparable right to remain underdescribed while exercising authority over others.

Chapter 34

Privacy and Future Reinterpretation

Let $M_t(R)$ denote the interpretation of a record R at time t . Then

$$M_t(R) \neq M_{t+\Delta}(R)$$

[Proposed theorem] even when the record itself is unchanged. Privacy therefore protects not only present secrets, but future vulnerability to unknown classificatory regimes. A trace disclosed under one social ontology may acquire a different meaning later, under new norms, institutions, or analytic tools.

The chapter adds a temporal limit to the prior chapters on mosaics, contextual transfer, model capture, and underdescription. Because records can be reinterpreted and recombined indefinitely, privacy sometimes requires forgetting, expiration, and friction against perpetual reuse. Otherwise every preserved appearance remains permanently available for renewed suspicion. The same protection against future reinterpretation becomes concealment when expiration or forgetting rules erase records that are still needed to reconstruct or challenge exercises of power.

Part VII

Volume IV: Law as an Epistemic Machine — Part VII. Evidence Before Action

Chapter 35

The Architecture of Proof

Legal proof separates factual belief from legally warranted judgment. A tribunal may suspect, even strongly, without yet occupying the institutional position from which action is authorized. The chapter therefore models proof as an authorized evidentiary path rather than as naked confidence:

$$E \xrightarrow{\text{admit}} E^* \xrightarrow{\text{test}} \tilde{E} \xrightarrow{\text{weigh}} J \xrightarrow{\text{authorize}} A.$$

[Definition] Here raw material E is first admitted into the record as E^* , then adversarially tested into $\tilde{E}$, then weighed into judgment J , and only then converted into authorized action A . Each arrow is a distinct site of institutional discipline and a possible refusal point. Legal proof is not belief with more confidence added; it is confidence produced through this public chain.

Chapter 36

Burdens and Standards of Proof

Burdens and standards of proof compare probable cause, reasonable suspicion, preponderance of evidence, clear and convincing evidence, and proof beyond a reasonable doubt as different permissions to act under uncertainty. They are not merely different numerical thresholds, but an approximate threshold model clarifies their function:

$$A \text{ permitted if } P(H \mid E) > \theta_A.$$

[Toy model] The point is not that every doctrine secretly encodes a precise scalar θ_A . It is that each standard specifies how much evidentiary support must exist before some action A becomes licit. As the magnitude, irreversibility, and asymmetry of the contemplated intervention rise, the threshold should rise with them. Standards of proof are therefore calibrations of authorized action under uncertainty rather than meters of metaphysical certainty. And when a standard demands evidentiary movement no institution can realistically supply for the kind of harm at issue, the same rigor flips into an unmeetable burden that protects the uncertainty-producing status quo.

Chapter 37

Admissibility

Admissibility is treated here as a type system for institutional evidence. Material is not admitted merely because it exists or because it bears loosely on the facts; it must satisfy requirements of relevance, provenance, reliability, prejudice, and contestability. If $\mathcal{E}$ is the space of all available material, the admissibility operator is

$$A_c : \mathcal{E} \rightarrow \{0, 1, \perp\},$$

[Definition] where c is the procedural context, 1 means admitted, 0 refused, and $\perp$ means unresolved pending further foundation. The operator formalizes the thought that law does not ask only whether something is true. It asks whether that material is usable without deforming the path from appearance to institutionally warranted judgment. When those admissibility demands require forms of provenance or contestability that some injuries cannot realistically supply, the same rigor turns into an unmeetable burden that screens structurally important evidence out before it can be tested.

Chapter 38

Cross-Examination as Adversarial Testing

Cross-examination probes whether a claim survives transformations of perspective. A witness's assertion is not simply counted as one more datum; its dependence on memory, incentives, vantage, language, and prior interpretation is made contestable in public. The chapter captures this as a procedural robustness functional:

$$R(H) = \inf_{\delta \in \Delta} P(H \mid E + \delta),$$

[Definition] where Δ contains the admissible perturbations or challenges generated by questioning. A conclusion that collapses under minor, procedurally allowed challenges has low robustness even if it looked persuasive before testing. Cross-examination thus measures not bare confidence, but the stability of confidence under organized pressure.

Chapter 39

The Judge, the Jury, and Distributed Error

Judges and juries embody different error structures. A single expert decision-maker may produce consistency, but it also creates a concentrated failure point in which one bias or blind spot can propagate through the whole inferential chain. A group may diversify judgment while introducing conformity effects, polarization, and other collective pathologies. The chapter therefore studies Condorcet-style aggregation while refusing the simplifying fiction that juror errors are independent and identically distributed. The design problem is not to choose between pure expertise and pure democracy, but to distribute error so that no single recurrent failure mode becomes sovereign.

Chapter 40

Appeal and Reversible Judgment

Appeal is not redundant repetition. It changes the level at which error is inspected, often shifting from direct fact-finding to procedure, interpretation, or institutional jurisdiction. A decision system without appeal approximates an absorbing state:

$$J_t \rightarrow J_{t+1} = J_t.$$

[Toy model] Once judgment enters state J_t , the system supplies no authorized transition edge back out of it. Appeal reintroduces those edges into the state space. It allows some judgments to be revised without making every judgment permanently unstable, and thereby distinguishes present authority from irrevocable epistemic finality.

Part VIII

Volume IV: Law as an Epistemic Machine — Part VIII. Legal Friction as Freedom

Chapter 41

The Inconvenience Principle

The inconvenience principle states the central proposition of the monograph: where information is incomplete, actors are fallible, and coercive actions are asymmetrically costly, procedural inconvenience can enlarge substantive freedom. Its technical handle is a friction parameter ϕ , which delays intervention just enough to force justification before power hardens into fact. Social loss can then be modeled by

$$L(\phi) = L_{\text{false action}}(\phi) + L_{\text{missed action}}(\phi) + L_{\text{delay}}(\phi).$$

[Toy model] Too little friction produces impulsive coercion; too much blocks legitimate protection. The task of legal design is therefore not to maximize friction, but to locate it where epistemic uncertainty meets irreversible power. Friction changes sign and becomes attrition when the waiting it imposes does no evidentiary work, falls asymmetrically on those already exposed to power, and offers no clear route to completion.

Chapter 42

Warrants and the Cost of Looking

A warrant makes intrusion costly on purpose. Before officials may look, they must specify what they expect to find and why, converting generalized suspicion into a contestable claim that another institutional actor can inspect. The chapter represents a warrant as a bounded search authorization

$$W = (T, S, Q, \Delta t),$$

[Definition] where T is the target, S the scope, Q the evidentiary justification, and Δt the temporal duration. Each coordinate matters: the target prevents rummaging, the scope prevents drift, the justification ties looking to reasons stated in advance, and the time bound prevents an open-ended license. A warrant is thus the price law places on observation so that successful discovery cannot retroactively justify the search that produced it. The same protection turns into concealment when demands for ex ante specificity are applied so rigidly that recurring, publicly legible patterns of hidden abuse never become sufficient to authorize any scrutiny at all.

Chapter 43

Refusal as a Positive Operation

Refusal is not the absence of decision. It is a positive legal operation that prevents an inadequately supported transition:

$$R : (E, A) \mapsto \neg(E \vdash A).$$

[Definition] The refusal operator R gives law a way to say “not yet,” “not by this route,” or “not on this evidence.” In the Spharepop distinction, that matters because active refusal is different from mere failure to continue: it preserves the burden on institutions to justify intrusion without conscripting the subject into producing the case. A legal system that cannot formalize refusal has only crude acceptance and rejection, and loses the intermediate protections on which procedural liberty depends.

Chapter 44

Due Process as Constrained Updating

Institutional belief has state Θ_t . Unrestricted reaction to new evidence E_t would permit

$$\Theta_{t+1} = \arg \min_{\Theta} L(E_t; \Theta),$$

[Toy model] even when E_t is narrow or unrepresentative. Due process adds a regularization term:

$$\Theta_{t+1} = \arg \min_{\Theta} [L(E_t; \Theta) + \lambda D(\Theta, \Theta_t) + \gamma C(\Theta)],$$

[Toy model] where D penalizes unjustified institutional drift and C encodes procedural constraints.

Due process does not merely preserve an established prior, since precedent itself can encode systematic error. Its deeper function is to constrain the update rule: new evidence may revise institutional judgment, but it cannot immediately rewrite the entire structure by which evidence is interpreted.

A useful technical analogue appears in parameter-efficient machine learning. In AFF-Net, unrestricted fine-tuning of a large pretrained model on scarce medical data produces feature drift: the model reorganizes broadly acquired representations around the accidental regularities of a small sample and can perform worse than a system permitted to update only through constrained adapters (see Chapter 86 for the original technical context, and Chapter 74 for the full development of the update-budget formalism this chapter introduces informally). Due process imposes an analogous restriction on institutional belief revision. An early recording, witness statement, or accusation may contain genuine information, but it is not statistically entitled to overwrite the entire interpretive structure through which responsibility is assigned. Standards of evidence, cross-examination, precedent, disclosure, and appeal force the new signal to survive contact with a larger body of accumulated constraints. Their value is not that the inherited structure is infallible. It is that limited evidence should initially produce limited revision. The legal equivalent of feature drift occurs when a vivid but narrow sample reorganizes judgment before its provenance, representativeness, and causal context have been tested.

The permitted magnitude of an institutional update should be proportionate not to the vividness of the evidence, but to its tested evidentiary reach. The same constraint becomes immunity for bad priors when institutions treat inherited categories as presumptively self-vindicating and keep demanding ever more proof against patterns those categories were built not to see.

Chapter 45

Presumption of Innocence

The presumption of innocence is not an empirical prediction that most accused persons are innocent. It is an allocation rule governing who must bear the cost of uncertainty when coercive error is hardest to undo. If H_1 denotes guilt and H_0 innocence, and false conviction and false acquittal carry costs C_{FP} and C_{FN} , then the decision threshold reflects a normative weighting rather than probability alone:

$$\text{convict if } \frac{P(H_1 | E)}{P(H_0 | E)} > \frac{C_{FP}}{C_{FN}}.$$

[Toy model] The threshold therefore encodes an institutional judgment about asymmetric error costs. To presume innocence is to forbid accusation from counting as its own confirmation and to assign unresolved uncertainty away from coercive judgment until tested evidence overcomes that default. The presumption changes sign and becomes practical immunity when institutions let that default harden into a refusal to investigate, charge, or revise judgment even after tested evidence has accumulated.

Chapter 46

Delay, Liberty, and the Protected Interval

The period before judgment is not empty time but a protected interval in which competing histories remain institutionally possible. Let $\mathcal{H}_t$ be the admissible history set at time t . As evidence is tested, that set may shrink:

$$\mathcal{H}_{t+1} \subseteq \mathcal{H}_t.$$

[Definition] Legitimate judgment requires that the shrinkage occur through warranted exclusions—through admissibility, testing, burden, and authorized inference—rather than through the mere disappearance of unpopular alternatives under time pressure. Delay protects liberty when it preserves the interval in which those histories can still be publicly defended before action becomes irreversible.

Part IX

Volume V: Admissibility and Continuation —
Part IX. A General Theory of Institutional
State Change

Chapter 47

Record, Verify, Recognize, Authorize

The foundational chain of Record, Verify, Recognize, Authorize, and Continue is:

$$\text{Record} \not\Rightarrow \text{Verify} \not\Rightarrow \text{Recognize} \not\Rightarrow \text{Authorize} \not\Rightarrow \text{Continue}.$$

[Definition] where each operator names a distinct institutional stage rather than an entailment. Each term has its own formal role. Record creates persistence. Verification establishes specified relations between claim and trace. Recognition places the verified object within an institutional category. Authorization licenses an action. Continuation asks whether that authorization remains active across time.

The point of the chain is that no stage entails the next. A record may exist without verification; a verified trace may fail recognition; a recognized status may still not authorize coercion, payment, access, or speech; and an authorization may lapse, expire, or be interrupted rather than continue. Volume V therefore begins from typed non-implication: institutional state change is trustworthy only when these operations remain distinct enough to be separately inspected.

Chapter 48

Pop, Refuse, Bind, Collapse

The Spherepop primitives are translated here into legal and epistemic operations. Let P, R, B, C denote Pop, Refuse, Bind, and Collapse. Pop introduces a distinction into active consideration. Refuse blocks a proposed transition. Bind joins evidence, claim, person, rule, or responsibility under a relation. Collapse converts an unresolved field of alternatives into an operative determination.

The chapter then develops an operational algebra and asks which compositions commute:

$$C \circ B \stackrel{?}{=} B \circ C.$$

[Proposed theorem] the chapter's load-bearing claim is that these compositions generally fail to commute in legal settings. In most legal settings they do not. Binding evidence to a person before interpretive collapse differs from collapsing a narrative and only then assigning the person to it. The noncommutation matters because institutional consequence is order-sensitive: the same primitives can yield different public states depending on when refusal, binding, or collapse occurs.

Chapter 49

Merge, Link, and Unlink

This chapter develops the derived operations of merge, link, and unlink for evidentiary synthesis. To merge two files is stronger than to link them: merge fuses separate items into one operative object, while link preserves distinct records under a relation that can still be inspected. Unlinking a record from a claim does not erase the record, because the historical trace of association can remain even after the association is withdrawn.

That distinction supports a non-destructive model of correction. A false association can be revoked without pretending the historical association never occurred. The chapter therefore treats relational repair as a core institutional task: many large errors come not from missing facts, but from having linked, merged, or failed to unlink appearances in the wrong way.

Chapter 50

The Admissibility Boundary

Let Ω be the space of conceivable claims and $\mathcal{D}_c \subseteq \Omega$ the claims admissible in context c . The admissibility boundary is then

$$\partial\mathcal{D}_c,$$

[Definition] where Ω is the claim-space and c the forum context. which contains claims whose status is contested, incomplete, or dependent on further evidence.

Admissibility is therefore not identical with truth. It is a forum-specific judgment about which claims may presently bear consequence under given rules and capacities of interpretation. Much institutional conflict concerns motion of the boundary rather than direct disagreement about truth: actors fight over whether a claim counts as ready, legible, or procedurally fit for uptake at all.

Chapter 51

Continuation Geometry

A decision does not merely exist; it persists, expires, propagates, or is interrupted. Continuation geometry models that persistence as transport across institutional states:

$$T_\gamma : \mathcal{J}_x \rightarrow \mathcal{J}_y,$$

where γ is a procedural path and $\mathcal{J}_x$ is the space of judgments available at state x .

The central question is whether a judgment transported through different institutional paths remains equivalent. Amendments, stays, remands, transfers, substitutions, and delays do not simply consume time; they shape the path by which a judgment is carried. Continuation is thus geometric rather than merely chronological: sameness depends on the authorized route through the institutional field.

Chapter 52

Distinction Holonomy

When a claim travels through police, media, court, archive, and public memory, it may return to its apparent starting point with altered meaning. If transport around a loop γ changes the claim,

$$T_\gamma(J) \neq J,$$

the system exhibits distinction holonomy.

This names a precise form of path dependence. Repeated representation can transform an accusation even when every individual transmission appears locally defensible. The chapter's claim is not merely that institutions reinterpret, but that lawful circulation itself can accumulate residue: a distinction may remain formally continuous while the journey changes what the distinction now does.

Part X

Volume V: Admissibility and Continuation —
Part X. Public Institutions as Epistemic
Ecologies

Chapter 53

Separation of Powers

Separation of powers is treated here as structured non-identical observation. Different institutions possess different information, incentives, temporal horizons, and error modes, so they do not process appearances in the same way even when they address the same public matter.

The goal is not perfect independence, but prevention of correlated collapse. Institutional redundancy succeeds when disagreement remains possible long enough to expose hidden assumptions, local blind spots, or premature closures that a single integrated line of authority would normalize. The separation of powers is therefore an epistemic ecology before it is only a constitutional slogan.

Chapter 54

The Press and the Court

The press optimizes for timely public relevance; courts optimize, at least ideally, for procedurally warranted determination. Their conflict begins when public urgency is mistaken for legal sufficiency, or when legal restraint is mistaken for factual indifference.

This chapter compares their different loss functions and time horizons. Journalism widens the field of public salience quickly, often before a coercive forum is ready to bind consequences. Adjudication narrows that field through procedure, burden, and remedy. Keeping the two distinct preserves a productive tension between exposure and sanction rather than forcing either institution to counterfeit the other's task.

Chapter 55

Science and Adjudication

Scientific conclusions remain revisable, while legal decisions often require finite closure. The chapter therefore distinguishes replication, peer review, evidentiary hearings, precedent, and final judgment rather than treating science and adjudication as interchangeable truth machines.

Its central question is when scientific uncertainty is imported badly into law and when legal demands for binary resolution distort scientific evidence. Science can leave a matter open while accumulating better models; adjudication must often decide a case, allocate responsibility, and stop. The problem is not whether one institution should dominate the other, but how revisable inquiry is translated into deadline-bound public judgment.

Chapter 56

Archives and Institutional Memory

Archives preserve material beyond its present salience. Their democratic function is to keep future reconstruction possible, so the chapter models archival selection, retention, indexing, and degradation as active determinants of what a polity can later know.

An archive is not a warehouse but a future-facing admissibility structure:

$$\mathcal{A}_t \longrightarrow \{\text{questions answerable at } t + \Delta\}.$$

What is kept, discarded, misfiled, or allowed to decay changes the set of questions that later investigators, courts, journalists, or citizens can responsibly answer. Institutional memory is therefore a designed capacity, not a passive remainder.

Chapter 57

Bureaucracy and Slow Knowledge

Bureaucratic delay can conceal indifference or protect against haste. This chapter distinguishes pathological delay, capacity delay, verification delay, jurisdictional delay, and strategically imposed delay instead of treating slowness as a single phenomenon.

A normative theory of slow knowledge must ask what work occurs during the interval, who bears its costs, and whether the process remains inspectable. Delay is justified when it performs comparison, checking, routing, or error control that cannot be skipped without loss; it becomes suspect when time is consumed without epistemic work or when the costs of waiting are invisibly shifted onto those already exposed to institutional power.

Chapter 58

The Custodian Institution

The custodian institution does not dictate the final interpretation. It preserves records, boundaries, contestability, and routes of correction. Custodial authority is therefore distinct from sovereign authority.

This closing chapter reconnects Volume V to the broader custodian architecture. An institution is genuinely custodial when it maintains the conditions under which later agents can reconstruct why a decision was possible, refused, or reversed. What it keeps intact is not only data, but the traceability of admissibility, continuation, and correction across time.

Part XI

Volume VI: Adversaries, Platforms, and
Automated Judgment — Part XI.
Untrustworthy Actors

Chapter 59

Models of Adversarial Agency

Disciplined paranoia begins by replacing generic talk of “security” with explicit attacker models of increasing capability. The relevant question is not whether an institution faces adversaries in the abstract, but which forms of agency it can actually resist. The outline therefore orders adversarial power by a nested family

$$\mathcal{A}_0 \subset \mathcal{A}_1 \subset \cdots \subset \mathcal{A}_k.$$

[Definition] Nested adversary classes order hostile capability by increasing powers. Each step in the chain adds capacities an actor may possess: to lie, omit, frame, flood, intimidate, collude, impersonate, or exploit procedural rules.

Once these adversary classes are stated, suspicious appearances can be interpreted proportionately. The same surface anomaly may be consistent with error, routine incentive, or intentional manipulation; the institution should say whether its procedures are robust against, say, $\mathcal{A}_2$ but not $\mathcal{A}_5$, rather than pretending to a universal immunity it does not have. The chapter’s preface claim is therefore simple: disciplined suspicion starts with a bounded model of adversarial agency and asks which capabilities must be defeated before judgment can count as secure.

Chapter 60

Trust Networks and Correlated Capture

Trust should be evaluated over dependency graphs rather than by counting repetitions. The key danger is correlated capture: what looks like many confirming voices may in fact be many downstream copies of one compromised source. If reports $R_1, \dots, R_n$ share the same ancestor S , their agreement does not supply the evidentiary force that genuine independence would have supplied. [Toy model] The shared-ancestor sketch models apparent plurality as hidden dependence.

The outline's point is therefore structural rather than psychological. A thousand repetitions can remain only one underlying observation, one leak, one institution, or one gatekeeper echoed through many channels. Their joint evidentiary weight is then far smaller than it would be under independence, even if no single downstream speaker is consciously deceptive. Resilient trust networks must accordingly diversify origins, incentives, and review paths, because nominal plurality is useless wherever hidden dependence has already collapsed into correlated capture.

Chapter 61

Strategic Transparency

Strategic transparency treats publicity as a field of action rather than a passive condition. Actors can exploit demands for openness by staging visible compliance, overwhelming auditors with excessive volume, disclosing information in unusable form, or moving the consequential activity into categories that monitoring systems do not inspect.

Visibility therefore cannot be equated with innocence. A disclosure regime may be perfectly public and still fail epistemically if what becomes visible is selected for theater rather than audit. The chapter's preface thus marks transparency as strategic terrain: one must ask not simply what has been shown, but what has been made visible, what has been buried under excess, and what has been displaced into the blind spots of the monitoring architecture.

Chapter 62

Informants, Whistleblowers, and Anonymous Speech

Anonymity can protect corrective information while also reducing accountability. The outline therefore refuses both easy romanticism and easy suspicion. Anonymous speech matters because retaliation, surveillance, and captured channels often make attribution incompatible with disclosure; yet the loss of identity information shifts the burden of trust onto other features of the record.

The chapter treats anonymity as a redistribution of verification burdens. A nameless allegation is neither self-authenticating nor self-discrediting. What matters is whether the claim can be corroborated, whether documents or observations can be independently checked, and whether the surrounding pattern of evidence survives scrutiny without leaning on identity alone. Anonymous speech is thus a protective condition for some truths to surface, but never a substitute for verification.

Chapter 63

Manufactured Uncertainty

Manufactured uncertainty names a tactic by which power exploits skepticism itself. If every causal claim is described as too complex, too entangled, or too contestable ever to justify action, then accountability can be postponed indefinitely without anyone having to supply a better explanation.

The chapter distinguishes legitimate uncertainty from manufactured doubt by a practical criterion: whether the disputing actors clarify what evidence would resolve the issue. Honest uncertainty leaves open a path by which inquiry could discriminate among explanations. Manufactured uncertainty erodes that path, multiplying doubt while refusing to specify any resolving test. The preface claim is therefore that disciplined paranoia must be skeptical of skepticism when ambiguity functions not as intellectual caution but as a strategy for making judgment impossible.

Chapter 64

The Asymmetry of Defensive Paranoia

The same language of suspicion can have opposite political meaning depending on who is authorized to act. The less powerful may need defensive paranoia simply to survive institutions they cannot inspect, predict, or contest. The powerful, by contrast, may invoke the same vocabulary of vigilance to normalize surveillance over people who cannot resist it.

The asymmetry is not merely psychological; it is institutional. Suspicion from below often compensates for opacity and vulnerability, whereas suspicion from above can become an instrument of preemption, monitoring, and control. The chapter's preface therefore insists that paranoia cannot be evaluated in abstraction from power: identical epistemic postures differ morally and politically according to who bears the risk, who commands the apparatus of action, and who can answer back once suspicion is operationalized.

Part XII

Volume VI: Adversaries, Platforms, and Automated Judgment — Part XII. Platforms and Automated Publicity

Chapter 65

Platforms as Evidentiary Institutions

Platforms already function as evidentiary institutions. They collect records, rank claims, determine visibility, enforce categories, and preserve or erase histories long before any court, newsroom, or public inquiry confronts the material. What later counts as available evidence is therefore partly a consequence of platform architecture.

This role is quasi-legal even where it is not legally acknowledged. Decisions about retention, deletion, ranking, metadata, and categorical labeling shape chain of custody, salience, and retrievability, but the corresponding procedural safeguards are often thin or absent. The preface claim is accordingly that platforms do not merely host speech: they organize testimony, memory, and disappearance at scale, and so exercise evidentiary power that warrants institutional scrutiny.

Chapter 66

Moderation as Compressed Adjudication

At scale, moderation becomes compressed adjudication. Platforms cannot investigate each case as a court or full administrative process would; instead they substitute rules, classifiers, queues, and confidence thresholds for individualized inquiry. The governing problem is not whether error can be eliminated, but how judgment is allocated under constrained attention and asymmetric error costs.

This means that moderation architecture already decides which cases are seen first, which kinds of evidence count, which thresholds trigger action, and which mistakes are treated as tolerable. Compression is therefore substantive rather than merely logistical. The chapter's preface frames moderation as a system that formalizes public judgment under scarcity: procedural values are not absent, but translated into queue design, automated triage, and threshold-setting that decide at scale what full adjudication would decide in detail.

Chapter 67

Viral Evidence

Viral media generates attention faster than it generates verification. Let $V(t)$ denote visibility and $Q(t)$ verification quality. In the decisive early interval, the characteristic inequality is

$$\frac{dV}{dt} \gg \frac{dQ}{dt}.$$

[Toy model] The early-interval rate inequality models circulation outrunning verification. Visibility races upward while provenance, context, and corroboration mature much more slowly.

The consequence is that public judgment often reaches saturation before verification is complete. What millions have already seen, shared, and morally sorted acquires the force of settled evidence even when the record remains epistemically immature. The chapter therefore treats viral evidence as a timing problem: the issue is not that publicity is worthless, but that the rate of circulation can vastly exceed the rate at which evidentiary quality becomes fit for decisive judgment.

Chapter 68

The Permanent Record and the Mutable Person

A person changes while the record persists. The outline models this disparity as the difference between human change $X(t)$ and a frozen representation $R(t_0)$ retrieved again and again after its original moment. Digital systems can keep returning $R(t_0)$ to the present, thereby collapsing biographical time and encouraging institutions to treat a past snapshot as a standing portrait. [Toy model] The trajectory/archive pair models temporal misfit between persons and durable records.

The central question is when an accurate record becomes contextually false in continued use. $R(t_0)$ may remain true of what occurred at t_0 , yet misleading when used as if it fully represented the later person described by $X(t)$. The chapter's preface thus frames the permanent record not as a simple contest between truth and forgetting, but as a problem of temporal fit between durable archives and mutable persons.

Chapter 69

Algorithmic Accusation

Algorithmic accusation begins where systems convert irregularity into suspicion at scale. Risk scores, anomaly detectors, and related tools can legitimately flag cases for review, but only if institutions preserve the distinction between flagging and violation:

$$\text{Flag}(x) \not\Rightarrow \text{Violation}(x).$$

[Definition] The flag/violation gap fixes the difference between triage and adjudicated wrongdoing. A flag marks something as worthy of attention; it does not itself establish guilt, breach, or sanctionable wrongdoing.

The chapter's preface insists that automated systems become accusatory when they forget this non-implication. If a pipeline treats statistical deviation as though it were already a proved offense, then suspicion has been collapsed into judgment. The required discipline is therefore procedural as much as technical: algorithmic tools may prioritize review, but they must not silently convert scores, anomalies, or risk rankings into verdicts. When institutions preserve that separation only by demanding ever more proof without funding timely review, evidentiary rigor flips into an unmeetable burden that leaves automated suspicion in force without accountable resolution.

Chapter 70

Synthetic Evidence and Provenance Failure

Generative media raises the cost of verification, but it does not make all evidence worthless. The problem is better described as provenance failure: an environment in which artifacts can be fabricated, remixed, stripped of origin, and recirculated faster than ordinary trust practices can stabilize their history.

The chapter's technical inventory is consequently layered rather than magical. Cryptographic provenance, sensor authentication, chain of custody, watermarking, and model-based detection can each strengthen a record's status, while each also has limits that prevent any purely technical guarantee. Synthetic evidence therefore shifts emphasis from naive faith in appearances to auditable pathways by which an image, clip, or document acquires trustworthy lineage. The task is not to declare all media suspect, but to rebuild evidentiary confidence around provenance systems that remain contestable and incomplete. When institutions demand immaculate provenance even where authentic witnesses cannot supply it, evidentiary rigor flips into an unmeetable burden that excludes real evidence along with synthetic fraud.

Part XIII

Volume VII: Formal Synthesis — Part XIII. A Mathematics of Disciplined Distrust

Chapter 71

The Paranoia Functional

This chapter introduces the paranoia functional as a formal measure of disciplined distrust: a belief-state should be evaluated not only by fit with appearances, but by its sensitivity to alternative causal histories that remain live under the evidence E . The basic institutional measure is

$$\mathcal{P}(E) = \sum_{H \in \mathcal{H}} w(H) \mathbf{1}[H \text{ remains admissible under } E].$$

[Definition] Here $\mathcal{H}$ is the space of candidate histories, $w(H)$ weights them, and the indicator records whether E has genuinely ruled a history out. A high value of $\mathcal{P}(E)$ therefore need not mark confusion; it can mark justified refusal to collapse underdetermined cases too quickly.

A more refined version treats distrust through posterior dispersion rather than admissibility alone. The entropy-based variant is

$$\mathcal{P}_H(E) = - \sum_{H \in \mathcal{H}} P(H \mid E) \log P(H \mid E).$$

[Definition] This chapter's preface is thus simple: high entropy need not mean pathological uncertainty. Under evidentiary incompleteness, preserving structured multiplicity is part of epistemic responsibility, and $\mathcal{P}(E)$ and $\mathcal{P}_H(E)$ supply the formal objects on which the next chapters will define loss, reach, and bounded revision.

Chapter 72

Premature-Collapse Loss

This chapter defines the cost of committing too early to a single institutional story. Let τ be the time of commitment, let $\hat{H}_\tau$ be the history endorsed at that time, and let H^* be the eventual best-supported history. The premature-collapse loss is

$$L_{\text{collapse}}(\tau) = C(\hat{H}_\tau, H^*) + \alpha A(\tau) + \beta R(\tau),$$

[Toy model] where C measures inferential error, A the harm of actions authorized by that error, and R the cost of reversing an entrenched judgment once institutions have reorganized themselves around it.

The point is not that every delay is good. It is that haste creates a compound loss: a mistaken inference, the downstream actions taken under false confidence, and the additional burden of undoing a judgment once it has sedimented into records, procedures, or force. By writing that structure explicitly as $L_{\text{collapse}}(\tau)$, the chapter prepares the next step: determining how far any body of evidence may legitimately travel before it licenses this kind of costly closure.

Chapter 73

Evidentiary Reach

This chapter formalizes evidentiary reach as the bounded scope within which a datum materially changes warranted belief. For evidence E , the reach set is

$$\mathcal{R}(E) = \left\{ H : \inf_{M \in \mathfrak{M}} |P_M(H \mid E) - P_M(H)| > \epsilon \right\}.$$

[Definition] A proposition H belongs to $\mathcal{R}(E)$ only if, across the class $\mathfrak{M}$ of robustly admissible models, conditioning on E changes its likelihood by more than ϵ . Reach is therefore not vividness, nor mere rhetorical force; it is tested inferential movement that survives model variation.

The distinction matters because evidence can strongly bear on one claim while failing to justify transfers into broader narratives, character judgments, or policy conclusions. This chapter thus gives formal content to the rule that the magnitude of an update should track tested evidentiary reach rather than the emotional intensity of a record. Once $\mathcal{R}(E)$ is defined, the next chapter can bound not only where evidence reaches, but how much motion it may authorize.

Chapter 74

Update Budgets

This chapter’s thesis is that every responsible revision of belief should spend from an update budget whose size depends on the quality and reach of the incoming evidence. Drawing on the preceding account of premature-collapse loss and evidentiary reach, it proposes an informal rule: the better the provenance, independence, replication, representativeness, and adversarial survival of a record, the more movement it may justify. The budget idea clarifies why due process is not an ornamental delay but a mechanism for rationing how violently institutions may redraw a person’s standing.

$$\Theta_{t+1}^{\text{free}} = rg \min_{\Theta} L(E_t; \Theta).$$

[Toy model] An unconstrained institution would let new evidence drive whatever revision most reduces its local loss, regardless of how narrow or unrepresentative that evidence is. It also explains why weak evidence can trigger attention without licensing wholesale reinterpretation of a case, a citizen, or a threat environment.

$$D(\Theta_{t+1}, \Theta_t) \leq B(E_t).$$

[Proposed theorem] Here D measures revision size and $B(E_t)$ grows only with the tested quality and reach of the incoming record. But the same rigor becomes an unmeetable burden when institutions define $B(E_t)$ so parsimoniously for disfavored claimants that no attainable record can authorize meaningful correction. With revision now bounded, the next chapter can ask how privacy and publicity jointly shape the very evidence on which those budgets depend.

Chapter 75

Privacy–Publicity Duality

This chapter treats privacy–publicity duality as a joint constraint on legitimate institutional life. Let x denote personal information and g institutional action. Privacy constrains observation:

$$\mathcal{O}(x) \leq \kappa.$$

[Toy model] Privacy constrains observation by limiting personal observability. Publicity constrains hidden authorization:

$$\text{Opacity}(g) \leq \rho.$$

[Toy model] The publicity constraint limits how much official action may remain shielded from contestation. The first inequality limits how much of a person’s life may be rendered observable; the second limits how much official action may remain shielded from contestation.

A legitimate order must satisfy both constraints together. Maximum privacy can degenerate into concealed private domination, while maximum publicity can degenerate into universal exposure. The chapter’s core claim is therefore dual rather than scalar: freedom occupies neither endpoint. Reliable evidence depends on protected zones in which appearances do not instantly harden into public dossier, and legitimate power depends on publicity sufficient to prevent hidden authorization from becoming unanswerable rule.

Chapter 76

The Freedom Region

This chapter defines the freedom region as a feasible institutional set rather than a single score. The region is

$$\mathfrak{F} = \{(\kappa, \rho, \phi, \theta) : L_{\text{domination}} + L_{\text{exposure}} + L_{\text{error}} + L_{\text{inaction}} \leq \varepsilon\}.$$

[Toy model] The parameters collect jointly maintained constraints, while the losses register distinct failure modes: domination, exposure, error, and inaction. A society counts as free, on this view, when it remains inside the feasible region rather than at one privileged numerical optimum.

The importance of $\mathfrak{F}$ is that it refuses to collapse liberty into a commensurable scalar. Different orders may occupy different points of the region while preserving the same basic condition: persons are not forced into exposure, domination, or reckless institutional response. Freedom is thus modeled as a region of viable joint constraints, completing Part XIII's claim that disciplined distrust exists to protect a structured social space rather than to maximize suspicion or transparency in isolation.

Part XIV

Volume VII: Formal Synthesis — Part XIV. Dynamics, Geometry, and Holonomy

Chapter 77

Belief-State Manifolds

Part XIV begins by modeling institutional states as points on a belief-state manifold $\mathcal{M}$. In that picture, evidence produces local motion, but procedural and legal rules determine which directions are actually available. Not every imaginable revision is institutionally admissible; rather,

$$\dot{\Theta}_t \in \mathcal{D}_{\Theta_t} \subseteq T_{\Theta_t}\mathcal{M}.$$

[Definition] Here $T_{\Theta_t}\mathcal{M}$ is the tangent space at the current state Θ_t , and $\mathcal{D}$ is a distribution encoding evidentiary and procedural constraints on motion.

The geometric point is not decorative. Once belief is placed on $\mathcal{M}$, one can distinguish small local adjustments from prohibited or normatively dangerous transitions, and one can describe institutional learning as travel through a constrained space rather than as isolated proposition swaps. This chapter therefore provides the formal setting for the later claims about path dependence, curvature, and holonomy: the update problem is always a problem of constrained motion on a socially structured manifold.

Chapter 78

Nonintegrability and Institutional Path Dependence

Given the constrained distribution $\mathcal{D} \subseteq T\mathcal{M}$, this chapter argues that institutional revision is often nonintegrable: the order of admissible operations matters. Hearing a public accusation before reviewing exculpatory material need not lead to the same endpoint as encountering the same materials in reverse order, even when the final dossier contains identical items. Sequence changes salience, credibility, procedural posture, and the meaning later evidence can acquire.

That claim gives formal expression to institutional path dependence. When $\mathcal{D}$ is nonintegrable, there is no path-independent summary of inquiry; the route itself leaves residue in the final state. The chapter links this structure to narrative hysteresis and to distinction holonomy: after traveling around a procedural loop, the institution may return to what looks like the same question while carrying a different classificatory orientation. With nonintegrability in place, the next chapter can represent that order-sensitivity geometrically through curvature.

Chapter 79

Curvature of Reinterpretation

This chapter makes order-sensitive judgment explicit through the curvature of reinterpretation. For update operators ∇_i and ∇_j , curvature is registered by the commutator

$$[\nabla_i, \nabla_j]J = \nabla_i \nabla_j J - \nabla_j \nabla_i J.$$

[Definition] A nonzero value means that applying two evidentiary or interpretive updates in opposite orders yields different resulting judgment J . Sequence, framing, and procedural posture are therefore not external accidents; they are formally encoded in the geometry of revision.

The chapter's claim is modest but sharp. When the commutator does not vanish, reinterpretation is locally curved rather than flat: nearby judgments do not translate rigidly under new evidence, but bend differently depending on the route by which the evidence arrives. Legal and institutional order-sensitivity can thus be represented geometrically, setting up the field-theoretic account of public interpretation that follows.

Chapter 80

RSVP Fields of Public Interpretation

This chapter cautiously adapts the RSVP vocabulary into a field description of public interpretation. A scalar field Φ represents institutional potential or concentrated authority; a vector field $\mathbf{v}$ represents the directed flow of attention and evidence; and an entropy field S represents unresolved interpretive multiplicity. These are not metaphysical substances, but formal devices for describing how publicity channels evidence through an uneven environment.

The governing intuition is that evidence does not enter a neutral public space. It is transported through varying concentrations of authority, directional flows of salience, and pockets of unresolved or suppressed multiplicity. The chapter therefore treats public interpretation as field-mediated rather than frictionless: what can be seen, amplified, stabilized, or contested depends on the shape of Φ , $\mathbf{v}$, and S across the social space.

Chapter 81

Constraint Relaxation

This chapter studies how crisis loosens the regularizers that ordinarily discipline institutional revision. Let $\lambda(t)$ denote procedural regularization. In emergency conditions, $\lambda(t)$ is reduced, permitting larger updates from limited evidence and relaxing the ordinary barriers that would otherwise constrain motion across belief space. Constraint relaxation can therefore reveal neglected possibilities, but it can also authorize exactly the sort of overreach the earlier chapters were built to resist.

The formal issue is when temporary relaxation becomes durable feature drift at the scale of the state. Lowering $\lambda(t)$ may be justified when existing procedures are too rigid to register danger or correction, yet the same reduction enlarges the blast radius of weak evidence and weakens the protections embodied in bounded reach, update budgets, and the freedom region. This chapter marks that boundary: disciplined distrust sometimes requires slack, but only on terms that remain reversible and answerable.

Chapter 82

Reknotting After Error

This chapter treats correction after institutional falsehood as a structural repair problem. Once a system has collapsed onto a mistaken narrative, replacing one proposition is not enough: records, reputations, precedents, and downstream decisions have already been tied to the error. Reknotting names the needed response, modeled as a constrained transformation of the dependency graph that preserves valid subsequent relations while removing dependence upon the false judgment.

The image matters because institutional error propagates through linked authorizations rather than remaining local. Repair must therefore detach what was wrongly downstreamed from the bad judgment without destroying later relations that remain sound. Reknotting is neither simple restoration nor total reset; it is a publicly constrained reweaving of dependencies so that the institution can remember the error, undo its illegitimate inheritances, and continue without preserving the false knot at its core.

Part XV

Volume VIII: Applications and Case
Structures — Part XV. Domains of Difficult
Judgment

Chapter 83

Criminal Investigation

Criminal investigation is the clearest ordinary setting in which suspicion is both necessary and morally dangerous. This chapter applies the framework to eyewitness testimony, forensic evidence, confessions, surveillance footage, prosecutorial disclosure, and wrongful conviction. Its emphasis is structural rather than sensational: early hypotheses do not merely interpret evidence after the fact; they help determine which witnesses are re-interviewed, which traces are tested, which videos are reviewed, and which inconsistencies are ignored.

The chapter therefore treats investigation as a sequence of constrained updates rather than a march from clue to certainty. It asks how lineup practice, forensic testimony, confession procedure, video interpretation, and disclosure obligations should be organized when the gravest miscarriages often arise from institutions that mistake a guided search for a neutral discovery process.

Chapter 84

Public Disorder and Mass Recording

Events involving thousands of participants and millions of recordings provide the clearest example of evidence abundance without rapid comprehension. This chapter studies public disorder through multi-camera reconstruction, identity uncertainty, decentralized action, provocation, and temporal fragmentation. The central problem is that recordings proliferate faster than they can be synchronized, authenticated, and interpreted, so visibility expands more quickly than understanding.

The task, then, is not to deny what cameras reveal but to specify what each recording can bear. A clip may establish presence while leaving role, sequence, intent, and instigation unresolved; a large archive may promise total legibility while remaining practically unmastered in real time. The chapter uses this setting to show why contested crowd events require methods of reconstruction that match the scale and fragmentation of the evidence itself.

Chapter 85

War, Intelligence, and Atrocity Evidence

War pushes the theory into environments where evidence is urgent, partial, and strategically corrupted. This chapter expands the analysis to damaged archives, propaganda, classified sources, satellite imagery, chain-of-command inference, and delayed forensic access. It asks how institutions should reason when early reports may be morally urgent enough to guide warning or protection, yet still evidentially incomplete for public attribution, sanction, or prosecution.

The point is not that wartime judgment must wait for impossible purity, but that different decisions authorize different evidentiary travel. Classified intercepts, refugee testimony, crater analysis, command patterns, and later exhumations do not arrive in one unified form, and they rarely mature at the same speed. The chapter therefore treats atrocity assessment as a problem of tiered judgment under hostile conditions rather than as a choice between gullibility and paralysis.

Chapter 86

Medicine and Scarce Data

In medicine, good judgment often depends on learning carefully from data too scarce, noisy, and heterogeneous to bear the weight we want to place on it. After the previous chapter's treatment of high-stakes public evidence, this one turns to the clinic and laboratory, where rare conditions, small cohorts, noisy imaging, and shifting baselines make overconfident updating especially attractive and especially costly. It will use medical-imaging and model-training examples to show why constrained forms of revision can outperform wholesale retraining when evidence is limited: the problem is not merely lack of data, but lack of permission to generalize freely from it. The chapter then connects bedside decision, trial design, and algorithmic assistance under a single norm of epistemic thrift, asking what clinicians may infer, what they must bracket, and how uncertainty should be communicated to patients. Its larger point is that skepticism here is not therapeutic defeatism but a method for preserving signal when the environment punishes exuberant inference.

Chapter 87

Child Protection and Family Privacy

Child protection makes the privacy–publicity conflict especially severe. Privacy can shield intimacy and family autonomy, but it can also conceal abuse; publicity can surface danger, but indiscriminate exposure can produce irreversible harm. This chapter studies asymmetric evidence in family settings through mandatory reporting, confidential testimony, clinical observation, school-based warning signs, and bounded disclosure.

Its focus is on how weak, unevenly distributed signals should be combined without converting every suspicion into a license for invasive intervention. The chapter therefore asks how reporting duties, confidential channels, case review, and disclosure limits can be designed so that children are not trapped inside opaque authority while families are not destroyed by the uncontrolled spread of untested accusation.

Chapter 88

Employment and Institutional Reputation

Workplace investigations often proceed under incomplete evidence, uneven power, and intense pressure for rapid resolution. This chapter examines employment, academic, and professional settings in which allegations of harassment, retaliation, discrimination, incompetence, or disloyalty are filtered through managerial risk, confidentiality rules, and institutional reputation. The key distinction is between protective interim action and premature final judgment.

That distinction matters because organizations often need to reassign duties, separate parties, or pause access before they can know enough to impose lasting sanction. The chapter asks when such interim measures are justified, when reputational or career consequences outrun the evidence, and how institutions can avoid turning uncertainty into a self-protective narrative that quietly settles the case before any genuine testing occurs.

Chapter 89

Scientific Misconduct

Scientific misconduct is a domain in which evidence about the world cannot be separated from evidence about how the result was produced. This chapter takes up authorship disputes, data irregularities, peer review, replication failure, and institutional conflict in order to distinguish anomaly, suspicion, investigation, finding, and sanction. The distinction matters because communities often collapse these stages, treating every irregularity either as scandal already proved or as a problem too embarrassing to examine.

The chapter therefore asks how whistleblowing, image screening, data access, replication, and formal inquiry should interact when the object of suspicion is not a single proposition but an evidentiary process. Its aim is to show that disciplined paranoia is internal to science itself: trust in results depends on institutions that can investigate their own production conditions without confusing preliminary concern with final condemnation.

Chapter 90

Historical Reconstruction

Historical understanding matures over decades rather than hearings or news cycles. This chapter emphasizes the long temporal horizon over which archives open, witnesses speak, technical methods improve, and formerly peripheral facts become legible. The result is not a romance of eventual certainty, but a model of inquiry in which the evidentiary field itself changes as institutions declassify records, archaeological methods advance, and suppressed voices enter the archive.

On that horizon, gaps and silences are neither transparent proof nor excuses for invention. The chapter studies how historians move among newly opened files, late testimony, material remains, and revised interpretive frames, showing why responsible reconstruction must remain revisable for far longer than most legal or political judgments can wait.

Part XVI

Volume VIII: Applications and Case
Structures — Part XVI. Failure Modes of the
Theory

Chapter 91

When Procedure Protects Power

Due process is not equally available to all parties. Wealthy actors can convert evidentiary rigor into indefinite delay, procedural complexity, and reputation management, while poorer actors often experience rapid coercion under the same nominal system. This chapter addresses that asymmetry directly so that the book's defense of disciplined suspicion does not collapse into a romance of actually existing legal and administrative institutions.

The focus is on the failure mode in which procedure works exactly as designed for the protection of the already powerful. Rules about pleading, counsel, delay, discovery, confidentiality, and burden can be formally neutral while remaining usable only by those who can afford time, expertise, and retaliation risk. The chapter therefore asks when procedural friction serves truth and when it merely converts hierarchy into the appearance of fairness.

Chapter 92

When Privacy Conceals Domination

Privacy must not become an unreviewable jurisdiction. This chapter studies domestic, corporate, religious, military, and governmental secrecy as settings in which evidence can be prevented from entering public consideration at all. Its target is not privacy as such, but the conversion of opacity into a shield for domination by actors who control testimony, records, and exit.

The problem is structural: the more fully a domain can define its own confidentiality, discipline its own witnesses, and suppress its own documentation, the less likely it is that even serious abuse will become publicly legible. The chapter therefore asks how outside review, protected reporting, and limited disclosure can be introduced without pretending that every private sphere is simply another arm of the state.

Chapter 93

When Publicity Is Necessary

Some facts must be disclosed before adjudication is complete because delay itself creates continuing harm. This chapter derives the conditions under which provisional publication is justified without confusing publication with final judgment. It is concerned with cases in which private channels predictably bury evidence, isolate victims, or allow ongoing danger to persist while institutions wait for a completeness they have no intention of creating.

The issue, then, is not whether publicity is risky; it plainly is. The question is when investigative reporting, public warning, registries, or open documentation supply epistemic conditions that closed systems refuse to supply for themselves. The chapter argues that such publication can be warranted as a provisional, reviewable act when the alternative is not fairness but the durable invisibility of harm.

Chapter 94

Epistemic Nihilism

Causal complexity cannot imply that nothing is knowable. This chapter responds to the temptation to treat manipulation, conflict, and incomplete evidence as reasons to collapse every judgment into arbitrary interpretation. Its task is to establish levels of warranted confidence and to distinguish incomplete knowledge from the claim that disciplined comparison has become impossible.

That distinction matters because nihilism often presents itself as sophistication while quietly equalizing better and worse explanations. The chapter therefore asks how much can be responsibly inferred from convergent but imperfect records, what confidence levels different institutions may claim, and why the admission of uncertainty is not the same thing as surrendering the possibility of reasoned ranking.

Chapter 95

The Abuse of “More Context”

Requests for context can be legitimate, but they can also indefinitely postpone recognition of clear evidence. This chapter defines diminishing contextual returns and asks when further expansion no longer materially changes the conclusion relevant to the decision at hand. Its subject is the recurrent maneuver by which every supplied clarification produces a new demand for background, sequence, motive, culture, or comparative framing.

The point is not to defend decontextualized judgment. It is to distinguish context that discriminates among live explanations from context that merely delays accountability until witnesses scatter and attention fades. The chapter therefore offers a stopping rule: inquiry should keep widening only while the added frame could still alter the warranted threshold judgment, not simply because more surrounding information can always be imagined.

Chapter 96

The Abuse of Presumption

Presumptions allocate burdens, but they can harden into immunity. This chapter studies when protective defaults must yield to accumulated evidence and how that transition can occur without abandoning procedural restraint. Its focus is on the corruption that arises when presumptions meant to economize inquiry become status shields for officials, institutions, or favored narratives.

The guiding question is how a rebuttable presumption stays genuinely rebuttable. A defensible default may organize action while evidence is sparse; an abusive default keeps reallocating credibility upward even after contradictory material accumulates. The chapter therefore closes the volume by asking when evidence has become strong enough to dislodge a standing presumption without collapsing into the opposite mistake of treating every accusation as self-validating.

Part XVII

Volume IX: Institutional Design — Part XVII. Designing for Delayed Truth

Chapter 97

Reopenable Records

Institutions should preserve not only what was decided, but the provenance, uncertainty, dissent, and dependency structure that made the decision possible. Under conditions of delayed truth, a record that stores only its terminal conclusion invites later reviewers to mistake administrative closure for epistemic completion. A reopenable record therefore keeps track of excluded alternatives as well as accepted ones, and it states why those alternatives were set aside at the time.

The point is not to abolish decision, but to ensure that later evidence has somewhere responsible to attach. If provenance has been erased, if uncertainty has been flattened, if dissent has been dropped, or if dependencies among premises have been hidden, correction becomes needlessly difficult. Reopenable records are thus archives designed for revision: they preserve the materials by which institutional judgment can be updated without pretending that the first account was final merely because it was first.

Chapter 98

Layered Verdicts

Institutions should not force every inquiry into a binary of true or false when the evidentiary situation is more structured than that. They can instead issue typed determinations such as substantiated, unsubstantiated, disproven, unresolved, procedurally barred, or outside jurisdiction. A layered verdict says what kind of closure has been reached, at what level, and for what purpose.

The crucial requirement is that these statuses remain distinct. “Not proven” must not silently become either “false” or “probably true,” because that collapse lets administrative convenience rewrite epistemic meaning. A mature verdict architecture marks the difference between a claim that has been defeated, a claim that has not yet been established, and a claim the institution is not authorized or prepared to decide.

Chapter 99

Bounded Transparency

Transparency should be designed by reference to purpose, audience, timing, and contestability rather than treated as a choice between total secrecy and universal publication. The question is not whether to disclose everything, but how to disclose enough, to the right people, at the right time, and in a form that permits challenge without manufacturing needless exposure.

Between opacity and full publicity lies a whole repertoire of intermediate forms: delayed release, redaction, confidential review, selective disclosure, and independent escrow. These are not evasions of accountability, but design options for making scrutiny compatible with privacy, safety, and procedural fairness. Bounded transparency is therefore the architecture of answerability under conditions where disclosure itself can help or harm.

Chapter 100

Attention-Aware Archives

An archive fit for delayed truth must be designed for later recovery, not merely present storage. It should expose provenance graphs, uncertainty annotations, duplicate clusters, dissenting classifications, and discovery histories so that a later investigator can see how a record traveled, what was contested, what may have been overcounted, and what remained obscure.

Search is part of that architecture. If earlier salience models suppressed an item, the archive should still make it recoverable rather than ratifying the first ranking as destiny. Attention-aware archives therefore treat findability as a condition of accountability: what matters is not only whether information exists somewhere, but whether future inquiry can locate it despite the biases, omissions, and heuristics of earlier attention.

Chapter 101

Procedural Regularization

The adapter analogy now becomes an institutional design principle. High-impact decisions should be more strongly regularized: they deserve independent review, slower update schedules, and explicit rollback mechanisms because the cost of premature or badly grounded revision is high. Low-impact and reversible decisions can tolerate faster updating because their error costs are easier to unwind.

Procedural regularization thus scales with stakes and reversibility. The point is not to make every institutional process equally slow, but to ensure that consequential judgments cannot be reconfigured by narrow evidence, transient pressure, or local enthusiasm without passing through thicker constraints. Good design decides in advance where the system should resist drift and where it may safely adapt quickly.

Chapter 102

Rights as Computational Constraints

Rights do not merely evaluate institutional action after optimization has finished; they limit the search space of permissible action from the start. If an institution can technically do many things, rights specify which of those things it is allowed even to consider as eligible outputs.

If $\mathcal{A}$ is the set of technically possible actions, rights define

$$\mathcal{A}_{\text{legal}} = \{a \in \mathcal{A} : a \models \mathcal{R}\}.$$

Optimization must therefore occur inside $\mathcal{A}_{\text{legal}}$, not across $\mathcal{A}$ as a whole with rights added later as a moral gloss. On this view, rights are computational constraints on collection, inference, retention, and intervention—architectural limits on what the institution may treat as actionably available.

Chapter 103

Audit Without Total Surveillance

Accountability does not always require universal exposure of the personal data on which institutions act. A system can be audited through selective audit, cryptographic commitments, access logs, independent custodians, zero-knowledge methods, statistical inspection, and trigger-based disclosure. These mechanisms make it possible to test whether officials acted lawfully without making every subject fully visible.

The design aim is to separate oversight from indiscriminate surveillance. Audit should track who accessed what, when, under what authority, and with what downstream effect, while keeping ordinary persons from becoming permanently inspectable merely because institutions need to be checked. The result is a form of answerability that does not reproduce the panopticon in the name of control.

Chapter 104

Institutions That Can Say “Not Yet”

A mature institution must be able to say “not yet.” It must distinguish delay from refusal, refusal from denial, and provisional action from final judgment. Without those distinctions, every pause looks like evasion and every temporary measure risks masquerading as settled conclusion.

The chapter therefore proposes explicit procedural states for incomplete evidence together with deadlines for reconsideration. Such states let an institution acknowledge uncertainty without losing track of its duties: it can defer, preserve the question, protect against interim harm, and return on schedule. The competence to say “not yet” is neither paralysis nor indifference, but a disciplined way of refusing premature closure.

Part XVIII

Volume X: Normative Conclusions — Part XVIII. Freedom Under Causal Opacity

Chapter 105

Freedom as Protected Indeterminacy

Freedom includes the ability not to have one's social meaning exhaustively fixed by a partial record. A person should not become identical, for institutional purposes, with the first dossier, profile, or episode that happens to be available. Protected indeterminacy names the legal and political space in which incomplete descriptions are prevented from hardening into total identities.

This protection does not erase responsibility. It does not say that action is unknowable or that judgment is impossible. It says instead that responsibility must be assigned without pretending that any partial account captures the whole person. Freedom persists where institutions leave room for revision, surprise, and unfinished self-description.

Chapter 106

The Right to Contest the Model

When an institution acts through a representation $M(x)$, the represented person should ordinarily be able to know that the model exists, inspect the relevant inputs, challenge the associations it draws, introduce counterevidence, and seek revision. Model-governed power is not legitimate merely because it is technically sophisticated; it must remain contestable by the person it classifies.

The right to contest the model therefore reaches beyond a narrow complaint about outputs. It concerns the representational apparatus itself: what was treated as relevant, how signals were linked, which proxies were accepted, and by what procedure the model may be corrected. Where institutions govern through $M(x)$, contestability is part of what makes representation politically tolerable.

Chapter 107

Equality of Evidentiary Standing

Formal equality is inadequate when different actors possess radically unequal ability to collect, preserve, interpret, and present evidence. Equality of evidentiary standing concerns practical access to the operations by which institutional reality is produced: who can get a record made, who can keep it alive, who can have it believed, and who can force it into the path of decision.

The issue is therefore not simply whether everyone may speak, but whether everyone can participate in the evidentiary machinery that turns speech into institutional fact. A free order requires more than neutral rules on paper; it requires conditions under which persons and groups are not structurally disabled from making their evidence count.

Chapter 108

Publicity of Power, Privacy of Persons

The normative orientation of this entire work can be stated simply.

Institutions exercising coercive power should be increasingly inspectable as their power grows. Persons should not become increasingly exposed merely because institutions acquire greater capacity to observe them.

The asymmetry is justified because institutions claim authority over others, whereas persons do not ordinarily owe the public exhaustive legibility. Publicity should therefore rise toward power, while privacy protects the ordinary conditions of personhood from becoming raw material for ever more capable systems of observation.

Chapter 109

Disciplined Paranoia

Disciplined paranoia begins from five standing assumptions: every record may be incomplete, every observer situated, every model contestable, every institution fallible, and every concentration of power vulnerable to capture. Those assumptions widen inquiry by refusing to equate the current account with the whole truth.

But disciplined paranoia is not permission for coercion by hunch. It requires specific evidence before specific coercion. Its formula is

maximum permission to question + strict limits on acting from suspicion.

It is therefore a civic ethic of expansive doubt joined to severe restraint on what doubt alone may authorize.

Chapter 110

The Freedom Created by Law

The final chapter returns to the apparent absurdity with which the book began: why should an innocent person have to endure formal proof, hearings, and delay? Because no institution receives innocence or guilt directly from the world. It receives records, testimony, measurements, classifications, and arguments.

The inconvenience of law creates freedom when it prevents those representations from becoming coercive facts too quickly. Legal procedure is therefore not merely an obstacle placed between knowledge and action. It is part of the machinery by which knowledge capable of authorizing action is produced.

The book closes, then, with its governing formulation:

Paranoia is powerful because causal reality exceeds every available account of it. Privacy prevents the account from becoming total exposure. Publicity prevents authority from becoming total secrecy. Evidence prevents suspicion from becoming fact. Procedure prevents fact from becoming force without passing through contestable form. Freedom persists in the intervals maintained among these operations.

Part XIX

Mathematical Appendices

% Appendix A: Probability, Bayesian Updating, and Model Uncertainty % Status: prose-drafted (full text supplied)

Appendix A

Probability, Bayesian Updating, and Model Uncertainty

A.1 Purpose and scope

The argument of this book depends upon a distinction that ordinary discussions of evidence often obscure. Uncertainty may arise because the available observations do not determine which hypothesis is true, but it may also arise because the observer has not represented all the relevant hypotheses, does not know how observations were generated, cannot establish the independence of sources, or cannot determine whether the present case belongs to the population from which prior expectations were derived. These are mathematically different forms of uncertainty.

Probability theory is indispensable for reasoning under uncertainty, but it does not transform incomplete records into complete causal knowledge. A probability is always defined relative to some specification of possible outcomes, distinctions among hypotheses, information available to the reasoner, and assumptions about how evidence relates to those hypotheses. Bayesian updating can describe how belief should change inside such a specification. It cannot guarantee that the specification contains the correct causal history.

This appendix develops the probabilistic foundations needed for the main text. Its purpose is not to reduce legal judgment to numerical calculation. It is to clarify where calculation is possible, where judgment enters the calculation, and why procedural constraints remain necessary even when probabilistic reasoning is formally impeccable.

The governing distinction is between uncertainty within a model and uncertainty about the model:

$$\underbrace{P(H \mid E, M)}_{\text{uncertainty represented within model } M} \quad \text{and} \quad \underbrace{P(M \mid E)}_{\text{uncertainty concerning the model itself}} .$$

A further possibility lies outside both expressions:

$$M^* \notin \mathfrak{M},$$

where M^* is the causally appropriate model and $\mathfrak{M}$ is the family of models under consideration. In that case, neither a precise posterior within one model nor an average over the represented models captures the full extent of ignorance.

A.2 Probability spaces and represented possibilities

A probability space is a triple

$$(\Omega, \mathcal{F}, P),$$

where Ω is a sample space of possible outcomes, $\mathcal{F}$ is a collection of events to which probabilities may be assigned, and P is a probability measure satisfying

$$P(\Omega) = 1,$$

$$P(A) \geq 0 \quad \text{for every } A \in \mathcal{F},$$

and, for pairwise disjoint events $A_1, A_2, \dots$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

The first limitation appears before any probability is assigned. The space Ω must already distinguish the possibilities that matter. If Ω contains only the alternatives "intentional" and "accidental," then the analysis has no direct place for coercion, misunderstood instruction, distributed responsibility, machine failure, incapacity, staged evidence, or combinations of these conditions. They may be compressed into the two represented categories, but the resulting posterior cannot reveal that the categories themselves were inadequate.

Suppose an investigator considers three hypotheses:

$$\mathcal{H} = \{H_1, H_2, H_3\}.$$

These might represent intentional action, accidental action, and technical failure. The condition

$$P(H_1) + P(H_2) + P(H_3) = 1$$

does not mean that reality must belong to one of these categories. It means only that the probability model has treated them as exhaustive. If a fourth history H_4 is later discovered, the original normalization appears overconfident retrospectively because probability mass was distributed among an incomplete set.

This motivates a distinction between formal exhaustiveness and causal exhaustiveness. A partition is formally exhaustive when its cells cover the stipulated sample space. It is causally exhaustive only if the sample space adequately represents the ways in which the event could actually have occurred. Probability theory guarantees the former once the model is specified. It does not independently guarantee the latter.

The epistemic paranoia defended in this book begins at precisely this point. It asks not merely which represented hypothesis has the largest probability, but whether the represented alternatives deserve to be treated as exhaustive.

A.3 Random variables, observations, and records

A random variable is a measurable function

$$X : \Omega \rightarrow \mathcal{X}.$$

In the present context, X may denote a causally relevant but partially hidden condition, while another variable Y denotes what is observed. The observation is generated through some channel:

$$Y = \mathcal{O}(X, \eta),$$

where η includes noise, occlusion, measurement error, compression, selection, and uncontrolled variation.

The record available to an investigator may be a further transformation:

$$R = \mathcal{K}(Y, \zeta),$$

where $\mathcal{K}$ represents recording, storage, retrieval, editing, or transcription and ζ represents another source of distortion. The complete chain is therefore

$$X \longrightarrow Y \longrightarrow R \longrightarrow E,$$

where E is the portion of the record admitted or otherwise treated as evidence.

These variables should not be conflated. The probability of a record given a hypothesis,

$$P(R \mid H),$$

is not necessarily the probability of the underlying event given that hypothesis. Likewise, the probability that a record becomes visible to an investigator may depend upon both its content and the institutional pathway through which records are selected:

$$P(E \mid R, H) \neq P(E \mid R).$$

This inequality expresses informative selection. Evidence is not always a random sample of the available record. It may be selected because it is vivid, incriminating, easy to explain, compatible with an early theory, favored by a ranking algorithm, or supplied by a strategically interested actor.

The evidentiary process should consequently be represented as a generative system:

$$P(H, X, Y, R, E) = P(H)P(X \mid H)P(Y \mid X, H)P(R \mid Y, H)P(E \mid R, H).$$

Every conditional distribution introduces assumptions. The final posterior $P(H \mid E)$ inherits all of them.

A.4 Conditional probability

For events A and B with $P(B) > 0$, conditional probability is defined as

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

This represents the probability assigned to A after restricting attention to situations in which B occurs. It follows that

$$P(A \cap B) = P(A | B)P(B) = P(B | A)P(A).$$

Conditional probability is not causal direction. The expression

$$P(H | E)$$

concerns belief in H after observing E ; it does not state that E causes H , nor even that H directly causes E . Both may arise from common causes, and the evidence-selection process may itself depend upon the hypothesis under investigation.

An important source of error is the inverse fallacy:

$$P(E | H) \neq P(H | E).$$

A piece of evidence may be highly probable if a hypothesis is true without making that hypothesis highly probable given the evidence. The posterior also depends upon how commonly the evidence occurs under alternatives and upon the prior plausibility of the hypothesis.

If an unusual visual pattern occurs with probability 0.9 when manipulation has occurred but with probability 0.05 when no manipulation has occurred, the observation favors manipulation. Yet if manipulation is extremely rare in the relevant population, the posterior probability may remain modest.

Let

$$P(H) = 0.001, \quad P(E | H) = 0.9, \quad P(E | \neg H) = 0.05.$$

Then

$$P(E) = P(E | H)P(H) + P(E | \neg H)P(\neg H),$$

so

$$P(E) = 0.9(0.001) + 0.05(0.999) = 0.05085.$$

Therefore,

$$P(H | E) = \frac{0.9(0.001)}{0.05085} \approx 0.0177.$$

The evidence raises the probability of manipulation from approximately 0.1 percent to approximately 1.77 percent, a substantial relative increase but a low absolute probability. The distinction matters because public argument often describes any evidence that moves probability upward as though it established a high posterior confidence.

A.5 Bayes' theorem

Bayes' theorem follows directly from the symmetry of the joint probability:

$$P(H | E) = \frac{P(E | H)P(H)}{P(E)}.$$

For mutually exclusive and exhaustive hypotheses $H_1, \dots, H_n$,

$$P(H_i | E) = \frac{P(E | H_i)P(H_i)}{\sum_{j=1}^n P(E | H_j)P(H_j)}.$$

The four terms have distinct interpretations. The prior $P(H_i)$ represents belief before the new evidence is incorporated. The likelihood $P(E | H_i)$ describes how compatible the evidence is with the hypothesis. The marginal likelihood

$$P(E) = \sum_j P(E | H_j)P(H_j)$$

normalizes the posterior. The posterior $P(H_i | E)$ represents the updated state.

Bayesian updating is sometimes summarized as "begin with a prior and update on evidence." That formulation is correct but incomplete because it suppresses the work involved in determining what counts as evidence, how its generation should be modeled, which hypotheses should be compared, and whether the likelihoods are stable across contexts.

In a legal or historical investigation, the abstract expression

$$P(H | E)$$

may conceal a more realistic conditional structure:

$$P(H | E, S, R, C, T, M),$$

where S is the evidence-selection process, R its provenance, C the surrounding context, T its temporal relation to the event, and M the interpretive model. Two investigators can agree about Bayes' theorem while disagreeing profoundly about each of these conditioning variables.

A.6 Odds and likelihood ratios

For two hypotheses H_1 and H_0 , Bayes' theorem can be written in odds form:

$$\frac{P(H_1 | E)}{P(H_0 | E)} = \frac{P(H_1)}{P(H_0)} \cdot \frac{P(E | H_1)}{P(E | H_0)}.$$

Thus,

$$\text{posterior odds} = \text{prior odds} \times \text{likelihood ratio}.$$

The likelihood ratio

$$\Lambda(E) = \frac{P(E | H_1)}{P(E | H_0)}$$

measures how strongly E discriminates between the two hypotheses. If $\Lambda(E) > 1$, the evidence favors H_1 over H_0 . If $\Lambda(E) < 1$, it favors H_0 . If $\Lambda(E) = 1$, it does not discriminate between them.

This is more precise than saying that evidence is "consistent with" a hypothesis. Evidence can be perfectly consistent with H_1 while being equally or more consistent with H_0 . Mere compatibility is not probative strength.

Suppose a person runs from a location shortly after an incident. Let H_1 be the hypothesis that the person caused the incident and H_0 the hypothesis that the person did not. Even if

$$P(\text{running} \mid H_1) = 0.8,$$

the observation is weak if

$$P(\text{running} \mid H_0) = 0.7.$$

The likelihood ratio is

$$\Lambda = \frac{0.8}{0.7} \approx 1.14.$$

The behavior is more likely under guilt than innocence, but only slightly. Treating it as highly incriminating would confuse vividness with discrimination.

A likelihood ratio is also hypothesis-relative. Evidence that strongly distinguishes H_1 from H_0 may fail to distinguish H_1 from a third hypothesis H_2 . Binary presentation can therefore exaggerate evidentiary strength by excluding the alternative under which the evidence is most expected.

A.7 Sequential updating

Evidence usually arrives over time. If $E_1, \dots, E_n$ are observations, then

$$P(H \mid E_{1:n}) \propto P(H)P(E_{1:n} \mid H).$$

If the observations are conditionally independent given H ,

$$P(E_{1:n} \mid H) = \prod_{i=1}^n P(E_i \mid H),$$

and therefore

$$P(H \mid E_{1:n}) \propto P(H) \prod_{i=1}^n P(E_i \mid H).$$

In odds form,

$$O(H_1 : H_0 \mid E_{1:n}) = O(H_1 : H_0) \prod_{i=1}^n \Lambda(E_i).$$

Taking logarithms converts multiplication into addition:

$$\log O(H_1 : H_0 \mid E_{1:n}) = \log O(H_1 : H_0) + \sum_{i=1}^n \log \Lambda(E_i).$$

This representation is useful because repeated modest evidence can accumulate into strong support. It is also dangerous when purportedly separate evidence items share a common source. The product rule above requires conditional independence. When that assumption fails, multiplying individual likelihood ratios double-counts information.

Imagine that one original witness statement is repeated in twenty news reports. If the reports are treated as independent confirmations, the apparent evidentiary weight may grow exponentially. Yet all twenty reports may be descendants of one observation. Their dependency graph is

$$W \longrightarrow R_1, R_2, \dots, R_{20},$$

not twenty independent paths from the event.

Proper updating conditions on the shared source:

$$P(R_1, \dots, R_{20} \mid H, W)$$

rather than treating

$$P(R_1, \dots, R_{20} \mid H) = \prod_i P(R_i \mid H).$$

Repetition increases publicity. It does not necessarily increase independent evidence.

A.8 Source dependence and evidentiary genealogies

An evidence collection should be represented as a genealogy rather than a flat list. Let $G_E = (V_E, A_E)$ be a directed graph in which nodes are reports, recordings, measurements, or interpretations and edges indicate derivation, copying, citation, transformation, or common custody.

Two evidence items E_i and E_j are dependent when learning one changes the expected value of the other even after conditioning on the hypothesis:

$$P(E_i, E_j \mid H) \neq P(E_i \mid H)P(E_j \mid H).$$

Positive dependence is common when reports share a witness, sensor, database, editorial source, interpretive convention, or institutional incentive. Negative dependence can also occur, particularly when records compete for selection or when one interpretation suppresses another.

A crude approximation to the effective number of independent observations is

$$n_{\text{eff}} = \frac{n}{1 + (n-1)\rho},$$

where ρ is an average pairwise correlation. If $n = 100$ and $\rho = 0.1$, then

$$n_{\text{eff}} = \frac{100}{1 + 99(0.1)} \approx 9.17.$$

A hundred correlated observations may contain information comparable to roughly nine independent observations. This formula is not a universal evidentiary law, but it illustrates how quickly apparent abundance can collapse when dependency is recognized.

Provenance is therefore not auxiliary metadata. It determines the algebra by which evidence can legitimately accumulate.

A.9 Selection effects

Evidence that reaches an investigator is usually selected from a larger body of possible evidence. Let $S = 1$ mean that an item is selected or observed. The investigator commonly works with

$$P(H \mid E, S = 1),$$

not with the idealized quantity $P(H \mid E)$.

If selection depends upon both the hypothesis and evidentiary features, then

$$P(S = 1 \mid E, H)$$

must be included in the model. Bayes' theorem becomes

$$P(H \mid E, S = 1) \propto P(S = 1 \mid E, H)P(E \mid H)P(H).$$

Selection can produce a misleading association even when the underlying measurements are accurate. For example, suppose videos are uploaded primarily when events appear dramatic. The set of publicly circulating videos is then enriched for visually extreme behavior. An observer who treats it as a representative sample may infer that the extreme behavior is typical.

Selection also occurs during investigation. Once an initial hypothesis H_1 becomes dominant, investigators may preferentially retrieve evidence predicted by H_1 . Let Q_t be the query used at time t . If

$$Q_t = q(H_t),$$

then current belief influences which evidence becomes available for the next update:

$$H_t \longrightarrow Q_t \longrightarrow E_{t+1} \longrightarrow H_{t+1}.$$

This feedback loop can create confirmation without fabrication. Every retrieved item may be genuine, yet the retrieval process systematically excludes evidence expressed in an unexpected vocabulary, stored under an alternative category, or located in a part of the archive not searched by the prevailing theory.

A.10 Missing data

Let X be a variable of interest and R indicate whether it is observed. Classical statistical theory distinguishes several missingness mechanisms.

Data are missing completely at random when

$$P(R \mid X, Y) = P(R).$$

Missingness then does not depend upon observed or unobserved variables in the model. Data are missing at random when

$$P(R | X, Y) = P(R | Y),$$

so missingness may depend upon observed variables Y but not upon the missing value X after conditioning on Y .

Data are missing not at random when missingness continues to depend upon the unobserved value:

$$P(R | X, Y) \neq P(R | Y).$$

Legal, political, and historical records are frequently missing not at random. Actors may destroy incriminating documents, preserve exculpatory records, turn cameras on only after a confrontation begins, or withhold evidence because of shame, fear, loyalty, or institutional pressure. The absence of a record is therefore difficult to interpret.

It is nevertheless invalid to treat every absence as proof of concealment. Let D denote deliberate destruction and A the absence of a record. Although

$$P(A | D)$$

may be high, the posterior

$$P(D | A)$$

still depends upon the frequency of accidental loss, ordinary non-recording, retention policies, technical failure, and other explanations. Missingness should alter the uncertainty model without automatically resolving it.

A.11 Base rates and reference classes

A prior probability is often estimated from a reference class. Yet a case belongs simultaneously to many possible classes. An event may be classified by location, age, occupation, technology, institution, time period, behavioral pattern, or some conjunction of these attributes.

If the reference class is C , then the prior is

$$P(H | C).$$

Different plausible classes C_1 and C_2 can yield different priors:

$$P(H | C_1) \neq P(H | C_2).$$

This is the reference-class problem. It does not imply that priors are arbitrary. Some classifications are better supported, more causally relevant, or more stable than others. But prior selection is itself an inferential problem.

A broad class may provide stable statistics while ignoring local conditions. A narrow class may resemble the case closely while containing too few observations for reliable estimation. This creates a bias-variance trade-off:

$$\text{Expected error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible error}.$$

Institutional paranoia should therefore operate in both directions. It should resist the neglect of established base rates, but it should also resist treating a broad historical average as an infallible description of a particular case.

A.12 Priors as accumulated structure

A prior is sometimes portrayed as a subjective guess introduced before "the real evidence." This is misleading. Priors may encode extensive background knowledge, earlier data, physical constraints, institutional experience, or well-tested theory. A strong prior can prevent a small and noisy sample from producing a radical and unstable update.

Suppose a parameter θ has prior distribution

$$\theta \sim \mathcal{N}(\mu_0, \sigma_0^2)$$

and new observations satisfy

$$X_i \mid \theta \sim \mathcal{N}(\theta, \sigma^2).$$

After observing n values with mean $\bar{X}$, the posterior mean is

$$\mu_n = \frac{\sigma^{-2}n\bar{X} + \sigma_0^{-2}\mu_0}{\sigma^{-2}n + \sigma_0^{-2}}.$$

This can be written as a weighted average:

$$\mu_n = w_n \bar{X} + (1 - w_n)\mu_0,$$

where

$$w_n = \frac{n/\sigma^2}{n/\sigma^2 + 1/\sigma_0^2}.$$

When the new sample is small or noisy, w_n is small and the prior retains substantial influence. When the evidence becomes abundant and precise, w_n approaches 1.

This provides a mathematical analogue for constrained institutional updating. Precedent, evidentiary standards, adversarial testing, and procedural regularities can prevent a vivid but narrow sample from rewriting institutional belief immediately. Yet the analogy has limits. A prior can be badly chosen, historically biased, or protected from corrective evidence. Regularization is defensible only if sufficient evidence can eventually overcome it.

The proper principle is therefore not "preserve the prior," but:

$$\text{magnitude of revision} \propto \text{robust evidentiary reach}.$$

A.13 Posterior concentration and false confidence

Under favorable conditions, Bayesian posteriors concentrate around the true parameter as the amount of data grows. Informally,

$$P(\theta \in U(\theta^*) \mid E_{1:n}) \rightarrow 1 \quad \text{as } n \rightarrow \infty,$$

where θ^* is the true parameter and $U(\theta^*)$ is a neighborhood around it.

The phrase "under favorable conditions" carries much of the substantive burden. Posterior concentration may fail to produce truth when the observations are dependent, the model is misspecified, the data-generating process changes, the selection process is ignored, or the truth is not represented in the model family.

Under misspecification, the posterior may concentrate around the member of the model family closest to the true distribution according to some divergence, not around the true mechanism itself. Let P^* be the true distribution and $\{P_\theta\}$ the model family. The posterior may converge toward

$$\theta^\dagger = \arg \min_{\theta} D_{\text{KL}}(P^* \| P_\theta),$$

even though

$$P_{\theta^\dagger} \neq P^*.$$

More evidence can then produce greater confidence in the best available wrong model. This is one mathematical form of institutional feature drift: a system becomes increasingly certain because it has optimized its fit within an inadequate representational structure.

A.14 Model misspecification

Let a model M specify a family of distributions

$$\mathcal{P}_M = \{P_\theta : \theta \in \Theta\}.$$

The model is correctly specified if the data-generating distribution P^* belongs to $\mathcal{P}_M$. It is misspecified if

$$P^* \notin \mathcal{P}_M.$$

Misspecification can arise from omitted variables, incorrect functional form, unmodeled dependence, unstable measurement, selection bias, distribution shift, or an incomplete hypothesis space.

Suppose a model assumes

$$Y = \beta_0 + \beta_1 X + \varepsilon, \quad E[\varepsilon | X] = 0.$$

If an omitted variable Z affects both X and Y , then generally

$$E[\varepsilon | X] \neq 0.$$

The fitted association between X and Y may then absorb the influence of Z . Statistical precision around $\hat{\beta}_1$ does not repair the causal omission.

In evidentiary reasoning, an analogous error occurs when an observed action is interpreted without representing the institutional or interpersonal condition that generated both the action and its visibility. A narrow model can estimate its own parameters accurately while answering the wrong causal question.

A.15 Uncertainty within and between models

Suppose the investigator considers models $M_1, \dots, M_K$. Within-model uncertainty is represented by

$$P(H \mid E, M_k).$$

Between-model uncertainty is represented by

$$P(M_k \mid E).$$

Bayesian model averaging yields

$$P(H \mid E) = \sum_{k=1}^K P(H \mid E, M_k) P(M_k \mid E).$$

The posterior probability of a model is

$$P(M_k \mid E) = \frac{P(E \mid M_k) P(M_k)}{\sum_j P(E \mid M_j) P(M_j)},$$

where

$$P(E \mid M_k) = \int P(E \mid \theta_k, M_k) P(\theta_k \mid M_k) d\theta_k$$

is the model's marginal likelihood.

Model averaging is valuable because it prevents all uncertainty from being compressed into one favored specification. It still assumes, however, that the relevant alternatives are represented among $M_1, \dots, M_K$. The sum

$$\sum_{k=1}^K P(M_k \mid E) = 1$$

expresses completeness inside the analyst's model set, not proof that the model set exhausts causal reality.

A disciplined analysis should therefore distinguish three propositions:

$$P(H \mid E, M_k) \text{ is high,}$$

$$P(H \mid E) \text{ is high across represented models,}$$

and

H remains well supported under plausible unrepresented models.

Only the first two can be calculated directly from the stipulated family. The third requires robustness arguments, sensitivity analysis, domain knowledge, and institutional procedures that invite objections from outside the initial model.

A.16 Bayes factors and model comparison

For models M_1 and M_0 , the Bayes factor is

$$BF_{10} = \frac{P(E | M_1)}{P(E | M_0)}.$$

Posterior model odds are

$$\frac{P(M_1 | E)}{P(M_0 | E)} = BF_{10} \frac{P(M_1)}{P(M_0)}.$$

A Bayes factor compares two models; it does not establish that either is adequate. A large BF_{10} may mean that M_1 explains the evidence much better than M_0 while both remain poor descriptions of the underlying process.

This is especially important in adversarial public argument. A weak alternative can make a preferred account appear overwhelmingly strong. The relevant question is not merely whether H_1 defeats the account presented by its opponent, but whether the comparison set includes the strongest causally plausible alternatives.

The institutional importance of opposition, cross-examination, independent experts, and appeal partly lies in hypothesis-space expansion. An adversarial participant may contribute not just counterevidence but a model that changes the meaning of existing evidence.

A.17 Predictive adequacy and causal adequacy

A model can predict observations accurately while misrepresenting their causes. Predictive adequacy concerns the distribution

$$P(Y_{\text{new}} | E, M).$$

Causal adequacy concerns the behavior of the system under intervention:

$$P(Y | do(X = x), M).$$

These quantities need not agree. A variable may predict an outcome because it is a proxy for a hidden cause. Acting upon the proxy may then fail to change the outcome or may produce unjustified discrimination.

In institutional settings, predictive models frequently estimate

$$P(Y | X),$$

where Y is a future adverse event and X is a collection of personal features. Yet a decision to intervene based on X asks a different question:

$$P(Y | do(A = a), X).$$

The prediction that a person belongs to a high-risk category does not itself establish that a particular action against that person will be effective, proportionate, or justified.

The same distinction applies to narrative evidence. A behavioral pattern may predict involvement without explaining the event, and it may explain the event without establishing the legal responsibility required for sanction.

A.18 Calibration

A probabilistic forecaster is calibrated when events assigned probability p occur approximately a proportion p of the time. If q_i is the forecast for case i and $Y_i \in \{0, 1\}$ is the outcome, calibration informally requires

$$P(Y = 1 \mid q = p) = p.$$

For example, among a sufficiently large and relevant collection of cases assigned probability 0.7, approximately 70 percent should exhibit the predicted outcome.

Calibration is a population property. It does not determine the truth of an individual case. A well-calibrated system that assigns a person a 70 percent risk remains wrong in 30 percent of comparable cases. If an intervention is severe and irreversible, calibration alone cannot justify treating the probability as certainty.

Calibration can also vary across groups and environments. A predictor calibrated in population G_1 may be miscalibrated in G_2 :

$$P(Y = 1 \mid q = p, G_1) = p$$

while

$$P(Y = 1 \mid q = p, G_2) \neq p.$$

A system may appear calibrated overall while masking substantial local errors. Institutional use therefore requires subgroup analysis, temporal monitoring, and evaluation under distribution shift.

A.19 Discrimination, accuracy, and calibration

Three properties are often confused. Accuracy measures how often a categorical decision is correct. Discrimination measures how well a score separates cases with different outcomes. Calibration measures whether reported probabilities correspond to observed frequencies.

A model can be calibrated but weakly discriminative. If an event occurs in 10 percent of cases, assigning 0.1 to every case may be perfectly calibrated while providing no individual differentiation.

A model can discriminate well but be miscalibrated. It may rank high-risk cases above low-risk cases while systematically overstating the numerical probability.

A model can also achieve high accuracy in an imbalanced population by predicting the majority outcome in every case. If an adverse event occurs in 1 percent of cases, a system that always predicts "no event" is 99 percent accurate but useless for detection.

These distinctions matter whenever numerical performance is invoked to legitimate institutional action. "The system is accurate" does not specify what kind of accuracy, on which population, at which threshold, under what selection regime, and with what distribution of errors.

A.20 Proper scoring rules

A scoring rule evaluates probabilistic forecasts. For a binary outcome $Y \in \{0, 1\}$ and forecast $q \in [0, 1]$, the Brier score is

$$S_{\text{Brier}}(q, Y) = (q - Y)^2.$$

The logarithmic score is

$$S_{\log}(q, Y) = -[Y \log q + (1 - Y) \log(1 - q)].$$

A proper scoring rule encourages the forecaster to report its true probabilistic belief rather than strategically exaggerating or minimizing confidence.

Yet institutional actors do not always optimize for predictive accuracy. They may optimize for reputational safety, conviction rate, case closure, political responsiveness, audience engagement, or avoidance of visible false negatives. A formally well-designed evidentiary system can therefore be distorted by the loss function under which its participants operate.

Let the official loss be L_o and the actor's effective loss be L_a . If

$$L_a \neq L_o,$$

then behavior that appears irrational relative to official objectives may be rational relative to actual incentives. Institutional probability models must therefore include the ecology of incentives surrounding their production and use.

A.21 Credible intervals and confidence intervals

A Bayesian credible interval $C(E)$ for parameter θ satisfies

$$P(\theta \in C(E) \mid E) = 1 - \alpha.$$

This is a posterior probability statement conditional on the model and prior.

A frequentist confidence procedure produces intervals $C(X)$ such that, under repeated sampling,

$$P_{\theta}(\theta \in C(X)) = 1 - \alpha.$$

The parameter is treated as fixed and the interval as random. After observing one interval, classical frequentist theory does not generally interpret $1 - \alpha$ as the probability that the fixed parameter lies within that specific interval.

The distinction matters, but neither interval captures all relevant uncertainty. Both may omit measurement error, data selection, unmodeled dependence, specification uncertainty, and ambiguity about how the parameter maps onto the legal or historical proposition at issue.

A narrow interval can therefore coexist with broad causal uncertainty:

$$\text{Var}(\theta \mid E, M) \ll 1$$

while

$$\text{Uncertainty}(M, \mathcal{H}, S) \gg 0.$$

Numerical precision within a model should not be presented as exhaustive epistemic precision.

A.22 Imprecise probabilities and credal sets

When available information does not justify a unique probability distribution, uncertainty can be represented by a set of distributions:

$$\mathcal{P} = \{P_1, \dots, P_m\}$$

or, more generally, a convex set of admissible measures. Lower and upper probabilities are then

$$\underline{P}(A) = \inf_{P \in \mathcal{P}} P(A),$$

$$\overline{P}(A) = \sup_{P \in \mathcal{P}} P(A).$$

Instead of reporting

$$P(H \mid E) = 0.72,$$

one might report

$$P(H \mid E) \in [0.41, 0.81]$$

over a justified family of priors, likelihoods, source-dependence structures, and selection assumptions.

This interval is not a conventional confidence interval. It represents sensitivity to model uncertainty. A wide interval may be the most accurate statement available, even if institutions find it administratively inconvenient.

A decision can be called robustly authorized when every admissible model supports it:

$$\inf_{P \in \mathcal{P}} E_P[U(a, H)] > \sup_{P \in \mathcal{P}} E_P[U(a', H)]$$

for relevant alternatives a' . If model choice changes which action is preferred, the appropriate conclusion may be that the evidence does not yet warrant irreversible commitment.

Credal representations give formal content to disciplined paranoia. They preserve disagreement among plausible models without treating all imaginable models as equally credible.

A.23 Sensitivity analysis

Sensitivity analysis asks whether a conclusion survives reasonable variation in assumptions. Let the posterior depend upon parameters ψ representing priors, source reliability, dependence, selection, or likelihood specification:

$$P(H \mid E, \psi).$$

For an admissible assumption set Ψ , define the posterior range

$$\mathcal{I}_H(E) = \left[\inf_{\psi \in \Psi} P(H \mid E, \psi), \sup_{\psi \in \Psi} P(H \mid E, \psi) \right].$$

If the lower and upper bounds are close, the conclusion is relatively robust. If they span the operative decision threshold, the action depends upon unresolved modeling choices.

Local sensitivity examines derivatives such as

$$\frac{\partial P(H \mid E, \psi)}{\partial \psi_j}.$$

Global sensitivity varies several assumptions simultaneously. In causally dense settings, global sensitivity is often more informative because modeling choices interact.

A public claim should be regarded with particular caution when its apparent certainty depends on a narrow combination of assumptions that are omitted from its presentation. The question is not only "What is the estimate?" but "Which changes in the surrounding model would reverse it?"

A.24 Robust Bayesian updating

Suppose the prior belongs to a set $\mathcal{P}_0$, and the likelihood belongs to a set $\mathcal{L}$. Updating every admissible prior-likelihood pair produces a posterior set

$$\mathcal{P}_E = \{P(\cdot \mid E) : P \in \mathcal{P}_0, L \in \mathcal{L}\}.$$

A robust conclusion is one shared across $\mathcal{P}_E$. This makes explicit that evidence can be strong under one likelihood model and weak under another.

For example, the significance of a witness's identification depends upon assumptions about lighting, viewing duration, stress, prior familiarity, lineup construction, administrator behavior, memory contamination, and the population used to estimate error. Assigning a single likelihood ratio may obscure legitimate uncertainty about these conditions.

Robust updating does not mean refusing to conclude anything until every hypothetical model agrees. The admissible model family itself must be disciplined. Models should be included because they possess evidentiary, scientific, or causal justification, not merely because they preserve a desired conclusion.

Thus robust paranoia is bounded:

$$\mathfrak{M}_{\text{admissible}} \subsetneq \mathfrak{M}_{\text{imaginable}}.$$

A.25 Distribution shift

A model developed under distribution $P_{\text{train}}(X, Y)$ may be applied under a different distribution $P_{\text{target}}(X, Y)$. Distribution shift occurs when

$$P_{\text{train}}(X, Y) \neq P_{\text{target}}(X, Y).$$

Covariate shift occurs when

$$P_{\text{train}}(X) \neq P_{\text{target}}(X)$$

while the conditional relation $P(Y \mid X)$ remains stable. Label shift concerns changes in $P(Y)$. Concept drift occurs when

$$P_{\text{train}}(Y | X) \neq P_{\text{target}}(Y | X).$$

Historical and legal inference frequently involve all three. Technologies change, recording practices change, actors adapt to surveillance, categories acquire new meanings, and enforcement changes which cases enter the observed dataset.

A model that was once calibrated can become unreliable without any defect in its original construction. Continued authority therefore requires continued verification:

$$\text{Verified}_{t_0}(M) \not\Rightarrow \text{Verified}_{t_1}(M).$$

This is an instance of the distinction between authorization and continuation. A model's initial admissibility does not establish indefinite admissibility.

A.26 The value of additional information

Suppose an institution must choose an action $a \in \mathcal{A}$ under uncertainty about state H . Without additional evidence, it chooses

$$a^* = \arg \max_a E[U(a, H)].$$

If future evidence E' becomes available, the institution can condition its action on that evidence. The expected value of perfect information is

$$EVPI = E \left[\max_a U(a, H) \right] - \max_a E[U(a, H)].$$

The expected value of sample information is

$$EVSI = E_{E'} \left[\max_a E[U(a, H) | E'] \right] - \max_a E[U(a, H)].$$

These quantities formalize the benefit of waiting for additional evidence. Waiting is rational when the expected informational benefit exceeds the costs of delay:

$$EVSI > C_{\text{delay}} + C_{\text{investigation}}.$$

This condition depends strongly upon reversibility. If an immediate action is easily reversible, the cost of acting under uncertainty may be low. If the action produces imprisonment, public identification, permanent reputational damage, irreversible medical intervention, or destruction of records, the value of further information rises.

The protected interval defended in the main text is therefore not an unconditional preference for slowness. It is a period whose value depends upon expected learning and the cost of premature action.

A.27 Sequential probability tests and stopping

In sequential inference, evidence arrives over time and the investigator must decide when to stop collecting it. For hypotheses H_1 and H_0 , define the cumulative likelihood ratio

$$\Lambda_n = \prod_{i=1}^n \frac{P(E_i | H_1)}{P(E_i | H_0)}.$$

A sequential probability ratio test compares Λ_n with two thresholds. If

$$\Lambda_n \geq A,$$

one accepts H_1 ; if

$$\Lambda_n \leq B,$$

one accepts H_0 ; otherwise, sampling continues.

The mathematical elegance of this procedure depends upon assumptions rarely satisfied perfectly in public controversies. Evidence items may not be identically distributed, source dependence may be uncertain, hypotheses may evolve during the investigation, and the choice of which evidence to inspect next may depend upon current belief.

More importantly, institutional stopping is itself political. Investigations may stop because a threshold has been reached, because resources are exhausted, because public attention has moved elsewhere, because an authority prefers closure, or because further inquiry threatens an established institution.

A verdict at time τ therefore conditions not only on evidence $E_{1:\tau}$ but upon the stopping rule S :

$$P(H | E_{1:\tau}, S).$$

If stopping depends upon the observed evidence, ignoring S can exaggerate support. This is closely related to optional stopping, selective reporting, and the publication of analyses only when they reach a desired result.

A.28 Evidence accumulation under attention constraints

Suppose potentially relevant evidence items arrive at rate λ , while an institution can inspect items at rate μ . If $\lambda > \mu$, the unreviewed backlog grows on average:

$$\frac{dB(t)}{dt} = \lambda - \mu > 0.$$

Even when $\mu > \lambda$, prioritization may cause some categories to wait much longer than others. Let item i have salience score s_i , and suppose inspection priority is increasing in s_i . If legal relevance r_i is imperfectly correlated with salience, then high-salience low-relevance items may crowd out low-salience high-relevance items.

The probability that a relevant record is eventually recognized can be decomposed as

$$P(\text{recognized} | R) = P(\text{retrieved} | R)P(\text{understood} | \text{retrieved}, R)P(\text{escalated} | \text{understood}, R).$$

If the stages have probabilities p_1, p_2, p_3 , then

$$P(\text{recognized} | R) = p_1 p_2 p_3.$$

A record can be publicly accessible while having a very low probability of influencing institutional judgment. This establishes a probabilistic difference between publicity and effective evidentiary availability.

A.29 Saliency-biased sampling

Let the archive contain items $E_1, \dots, E_N$, each with legal relevance r_i and public salience s_i . A saliency-driven platform samples item i with probability

$$P(i \text{ displayed}) = \frac{\exp(\beta s_i)}{\sum_{j=1}^N \exp(\beta s_j)}.$$

As β increases, attention concentrates on the most salient items. If r_i and s_i are weakly correlated or negatively correlated, greater ranking efficiency with respect to saliency can reduce evidentiary quality.

The expected relevance of displayed evidence is

$$E[r] = \sum_{i=1}^N r_i \frac{\exp(\beta s_i)}{\sum_j \exp(\beta s_j)}.$$

There is no guarantee that

$$\frac{dE[r]}{d\beta} > 0.$$

A system can become increasingly effective at selecting what attracts attention while becoming less effective at selecting what supports accurate judgment.

This supplies a formal explanation for the paradox that intensive publicity may coexist with prolonged factual confusion.

A.30 Probability cascades

Suppose individuals announce beliefs sequentially. Person i receives a private signal S_i and observes the preceding public actions $A_1, \dots, A_{i-1}$. The rational posterior is

$$P(H \mid S_i, A_{1:i-1}).$$

Once enough earlier actors choose the same interpretation, their public actions may outweigh later private evidence. Subsequent actors then imitate the public sequence even when their own signals point elsewhere. An information cascade forms.

The cascade is fragile in principle because it may rest on little original evidence, but durable in practice because later conformity appears to provide additional confirmation. Observers may count the number of agreeing people without reconstructing the sequential dependency that produced their agreement.

Publicity can consequently create a distinction between the amount of visible consensus and the amount of independent information supporting it:

$$C_{\text{visible}} \gg I_{\text{independent}}.$$

Legal procedure attempts to interrupt such cascades by separating witnesses, excluding some prejudicial material, questioning source independence, and requiring that conclusions be justified through admissible evidence rather than through the mere social prevalence of a narrative.

A.31 Probability and standards of proof

It is tempting to identify legal standards of proof with numerical thresholds:

$$P(H \mid E) > \theta.$$

This representation is useful but incomplete. "Beyond a reasonable doubt" cannot be translated uncontroversially into a universal probability such as 0.90, 0.95, or 0.99. The posterior depends upon disputed priors, likelihoods, model choices, and the representation of alternative histories. Moreover, legal proof includes procedural provenance: how the evidence was obtained, tested, disclosed, and contested.

A purely numerical model can nevertheless clarify the allocation of error. Suppose action a_1 is conviction and a_0 is acquittal. Let C_{10} be the cost of convicting under innocence and C_{01} the cost of acquitting under guilt. Expected-loss minimization favors conviction when

$$P(H_1 \mid E)C_{01} > P(H_0 \mid E)C_{10},$$

or equivalently,

$$P(H_1 \mid E) > \frac{C_{10}}{C_{10} + C_{01}}.$$

If false conviction is assigned a much greater cost than false acquittal, the threshold rises.

But this derivation does not prove what the costs should be, how they should be compared, or whether they are commensurable. The threshold is a consequence of normative commitments, not a substitute for them.

The presumption of innocence is therefore better understood as a rule for allocating unresolved uncertainty than as a factual prior claiming accused persons are usually innocent. It states that uncertainty should not be converted into a particular form of coercion unless the burden of proof has been met.

A.32 Decision thresholds and reversible action

Not every action requires the same evidentiary threshold. Let a have severity $s(a)$, irreversibility $r(a)$, duration $d(a)$, and error asymmetry $c(a)$. A threshold function may be represented abstractly as

$$\theta(a) = f(s(a), r(a), d(a), c(a)),$$

with

$$\frac{\partial \theta}{\partial s} > 0, \quad \frac{\partial \theta}{\partial r} > 0, \quad \frac{\partial \theta}{\partial d} > 0$$

under ordinary assumptions.

A temporary and reviewable protective measure may therefore be justified at a lower level of confidence than a final punitive judgment. The difference must be encoded institutionally. Otherwise, provisional suspicion acquires permanent consequences while retaining the lower threshold appropriate only to temporary intervention.

Let a_p be a provisional action and a_f a final action. A legitimate system should ordinarily satisfy

$$\theta(a_p) < \theta(a_f)$$

and

$$\text{Review}(a_p, t^*) = 1$$

for a specified review time t^* . Without mandatory review, "provisional" becomes a rhetorical label attached to continuing punishment.

A.33 Action under ambiguity

Expected-utility theory assumes a probability distribution over states. Under severe model uncertainty, the institution may instead consider a family $\mathcal{P}$. Several decision rules are possible.

A maximin rule chooses

$$a^* = \arg \max_a \inf_{P \in \mathcal{P}} E_P[U(a, H)].$$

A minimax-regret rule chooses the action minimizing worst-case regret:

$$a^* = \arg \min_a \sup_{P \in \mathcal{P}} \left[\max_{a'} E_P[U(a', H)] - E_P[U(a, H)] \right].$$

An E -admissibility rule retains every action optimal under at least one admissible probability distribution. A robust-dominance rule authorizes a only if it outperforms alternatives under every admissible distribution.

These rules embody different attitudes toward ambiguity. None can be selected by probability theory alone. A legal system may use robust caution when individual liberty is threatened while accepting greater ambiguity for low-cost investigation. The asymmetry is normative and institutional.

A.34 Unknown hypotheses and residual probability

A closed hypothesis set assigns

$$\sum_{i=1}^n P(H_i) = 1.$$

To represent recognized incompleteness, one may introduce a residual category $H_\perp$:

$$\mathcal{H} = \{H_1, \dots, H_n, H_\perp\},$$

where $H_\perp$ denotes "some materially different unrepresented explanation." Then

$$P(H_{\perp}) > 0.$$

This does not solve the unknown-hypothesis problem, because a residual category lacks a detailed likelihood model. Nevertheless, it prevents the represented alternatives from receiving the entire probability mass automatically.

One may also use an open-world updating rule in which evidence that is poorly predicted by every represented hypothesis increases support for model expansion:

$$\max_i P(E \mid H_i) < \epsilon \implies \text{Expand}(\mathcal{H}).$$

The appropriate response to surprising evidence is not always to redistribute probability among existing narratives. It may be to introduce a new distinction.

This is the probabilistic analogue of Pop: the appearance of an inadequately represented possibility within the active hypothesis space.

A.35 Surprise, anomaly, and accusation

The surprisal of evidence E under model M is

$$I(E \mid M) = -\log P(E \mid M).$$

An observation with low predicted probability has high surprisal. But anomaly under a model is not proof of wrongdoing. It may indicate measurement error, distribution shift, omitted variables, poor model specification, rare innocent behavior, or intentional manipulation.

Thus,

$$\text{Anomalous}(E \mid M) \not\Rightarrow \text{False}(E)$$

and

$$\text{Anomalous}(E \mid M) \not\Rightarrow \text{Culpable}(H).$$

An anomaly detector identifies a mismatch between observation and expectation. It does not identify the cause of the mismatch. Institutions that convert anomaly scores directly into adverse action suppress the distinction between detection and adjudication.

The proper sequence is

$$\text{Anomaly} \rightarrow \text{Inquiry} \rightarrow \text{Causal differentiation} \rightarrow \text{Evidence} \rightarrow \text{Possible action}.$$

A.36 Reliability of testimony

Suppose a witness reports $R \in \{0, 1\}$ concerning proposition $H \in \{0, 1\}$. Define sensitivity and specificity as

$$s = P(R = 1 \mid H = 1),$$

$$c = P(R = 0 \mid H = 0).$$

Then

$$P(R = 1 \mid H = 0) = 1 - c.$$

The likelihood ratio associated with a positive report is

$$\Lambda_+ = \frac{s}{1 - c}.$$

This simplified model already requires empirical estimates of witness performance under relevant conditions. But testimony is not a fixed diagnostic instrument. Reliability can depend upon viewing conditions, delay, stress, expectations, questioning, confidence feedback, social identity, prior exposure, memory reconstruction, and incentives.

A more realistic specification is

$$P(R \mid H, Z),$$

where Z contains these conditions. If Z is incompletely observed, the investigator must integrate over its possible values:

$$P(R \mid H) = \int P(R \mid H, Z) dP(Z \mid H).$$

Multiple witnesses may share observation conditions or influence one another. Their reports should not be multiplied as independent likelihoods without a causal account of those dependencies.

Confidence is itself evidence only insofar as confidence is calibrated under comparable conditions. The statement "the witness is certain" describes a mental state. It does not automatically determine the accuracy of the proposition.

A.37 Reliability of recordings

A recording may fail through fabrication, alteration, truncation, misleading perspective, timestamp error, metadata loss, compression artifact, sensor malfunction, or contextual omission. Let A denote authenticity, I integrity, C contextual sufficiency, and R relevance. Probative adequacy might require all four:

$$Q = A \cap I \cap C \cap R.$$

Then

$$P(Q) = P(A)P(I \mid A)P(C \mid A, I)P(R \mid A, I, C).$$

The factors should not automatically be treated as independent. A recording can have perfect bit-level integrity while being contextually misleading. Cryptographic verification may establish that a file has not changed since a particular registration time; it does not establish what occurred outside the frame, who controlled the sensor before registration, or whether the displayed segment is representative.

A useful evidentiary statement should specify the level of verification:

$$\text{VerifiedHash}(R) \not\Rightarrow \text{VerifiedCapture}(R) \not\Rightarrow \text{VerifiedContext}(R) \not\Rightarrow \text{VerifiedInterpretation}(R).$$

A.38 Hierarchical evidence

Evidence often has nested structure. Videos belong to devices, devices to locations, observations to witnesses, witnesses to institutions, and institutions to broader political environments. A hierarchical model represents this dependence.

Suppose observation Y_{ij} is generated by source j within institution i :

$$Y_{ij} = \mu + \alpha_i + \beta_{ij} + \varepsilon_{ij},$$

where

$$\alpha_i \sim \mathcal{N}(0, \sigma_\alpha^2), \quad \beta_{ij} \sim \mathcal{N}(0, \sigma_\beta^2), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2).$$

The institutional effect α_i captures shared biases or conditions. Treating every Y_{ij} as independent ignores σ_α^2 and overstates the effective evidence.

Hierarchical modeling permits partial pooling. Information from related sources can inform one another without pretending that they are either completely independent or completely identical. This is particularly useful where evidence comes from multiple sensors, jurisdictions, laboratories, archives, or reporting organizations.

A.39 Latent-variable models

Many legally and historically relevant properties are not observed directly. Intent, knowledge, coercion, coordination, and responsibility are inferred from indicators.

Let Z be a latent state and $E_1, \dots, E_n$ observable indicators. A latent-variable model specifies

$$P(E_{1:n}, Z) = P(Z) \prod_i P(E_i | Z)$$

under conditional independence. But this assumption is substantive. Indicators may influence one another or share causes not represented by Z .

Latent categories can also acquire false concreteness. A fitted variable called "dangerousness," "credibility," or "intent" is not automatically identical to the legal or moral concept bearing that name. It is a mathematical construct inferred from selected indicators under a particular model.

The interpretation map

$$\iota : Z_{\text{model}} \rightarrow C_{\text{institutional}}$$

must therefore be justified separately. Statistical identifiability does not establish conceptual validity.

A.40 Exchangeability and its limits

A sequence $X_1, \dots, X_n$ is exchangeable if its joint distribution is invariant under permutations:

$$P(X_1 = x_1, \dots, X_n = x_n) = P(X_1 = x_{\pi(1)}, \dots, X_n = x_{\pi(n)})$$

for every permutation π .

Exchangeability permits observations to be treated as drawn from a common underlying process even when independence is not assumed directly. Much statistical learning relies, explicitly or implicitly, on some form of exchangeability between past and future cases.

Public events often violate it. Cases influence one another; actors learn from previous enforcement; recording technology changes; investigators inherit earlier categories; and publicity modifies subsequent behavior. The order is causally meaningful.

If a highly publicized case changes policy or behavior, later cases are not exchangeable with earlier ones. A model trained on the earlier distribution may therefore fail precisely because its deployment helped create a new distribution.

A.41 Censoring and survival of records

Records survive for different lengths of time. Let T denote time until deletion, degradation, loss, or inaccessibility. The survival function is

$$S(t) = P(T > t).$$

If records associated with certain actors, institutions, or event types have different survival distributions, the historical archive becomes systematically distorted.

Let Z denote record class. Then

$$S(t \mid Z = z_1) \neq S(t \mid Z = z_2).$$

Later investigators observe not the original universe of records but the subset that survived:

$$\mathcal{R}_t = \{R_i : T_i > t\}.$$

The probability that a historical claim is supportable may therefore depend upon preservation policy, material durability, classification, institutional continuity, and accidents of discovery. Absence from the surviving archive cannot be treated as simple evidence of historical absence.

A.42 Publication bias and the visible evidentiary record

Suppose a study or report is published with probability depending upon its result Z :

$$P(S = 1 \mid Z).$$

If striking, incriminating, statistically significant, or politically useful findings are more likely to become public, then the visible record has distribution

$$P(Z \mid S = 1) = \frac{P(S = 1 \mid Z)P(Z)}{P(S = 1)}.$$

This differs from the underlying distribution $P(Z)$.

The same mechanism applies beyond scientific publication. Allegations that fit established narratives may be reported more readily; exculpatory findings may appear later and receive less attention; routine non-events rarely generate records at all. Public evidence is consequently selected for reportability.

Transparency about selection mechanisms is as important as transparency about individual items. A complete database of selected reports is not a complete representation of the process from which those reports were selected.

A.43 Multiple comparisons and causal storytelling

When investigators search a large archive for patterns, some striking associations arise by chance. If m independent hypotheses are tested at significance level α , the probability of at least one false positive is

$$1 - (1 - \alpha)^m.$$

For $m = 100$ and $\alpha = 0.05$,

$$1 - (0.95)^{100} \approx 0.994.$$

The assumptions of independence and identical testing are simplified, but the general problem remains: a sufficiently large evidentiary archive contains coincidences.

Post hoc narrative construction can make these coincidences appear inevitable. Once a pattern has been found, the search process that generated it disappears from the presentation. The observer sees the selected alignment but not the thousands of alternatives inspected and discarded.

Probative assessment must therefore condition on the search procedure. A pattern predicted before inspection generally carries different evidentiary weight from one selected after extensive exploration.

A.44 Overfitting an event

Overfitting occurs when a model captures idiosyncrasies of observed data that do not generalize. In ordinary prediction, one compares training and test performance. In a unique historical or legal event, there may be no directly comparable test set.

Nevertheless, narrative overfitting can be defined as constructing an explanation with enough adjustable components to accommodate every known detail while providing little independent predictive or retrodictive constraint.

Let model complexity be $K(M)$ and fit be $L(E | M)$. A penalized criterion has the form

$$\text{Score}(M) = -\log P(E | M) + \lambda K(M).$$

A narrative that explains every fact only by introducing a new auxiliary assumption for each discrepancy may fit well while possessing poor explanatory economy.

Yet simplicity cannot serve as an absolute rule. Real causal histories can be complicated. Complexity penalties are justified because flexible models can imitate evidence easily, not because reality is guaranteed simple.

The proper concern is the relation between complexity and constraint: does the model take risks, exclude possibilities, and generate expectations that could prove it wrong?

A.45 Regularization and constrained updating

In statistical estimation, regularization limits how drastically a model changes to fit available data. A regularized objective is

$$\hat{\theta} = \arg \min_{\theta} [L(E; \theta) + \lambda \Omega(\theta)],$$

where L measures fit to evidence and Ω penalizes complexity or deviation from an established structure.

A proximity regularizer takes the form

$$\Omega(\theta) = \|\theta - \theta_0\|^2,$$

where θ_0 is the previous or pretrained state. Large λ resists rapid revision; small λ permits the new data to dominate.

The AFF-Net analogy belongs here. A broadly pretrained backbone contains structure learned from a large and diverse dataset. Full fine-tuning on scarce labeled medical data allows a small sample to alter the entire representation, potentially capturing spurious features. Freezing most of the backbone and training a limited adapter restricts the update to a smaller subspace.

An institutional analogue is

$$\Theta_{t+1} = \arg \min_{\Theta \in \mathcal{C}} [L(E_t; \Theta) + \lambda D(\Theta, \Theta_t)],$$

where $\mathcal{C}$ encodes procedural constraints. Early evidence may modify some beliefs without rewriting every prior classification, precedent, presumption, or standard of proof.

The analogy must not be extended into an identity. Legal institutions are not passive pretrained models; their inherited structures embody power and may require radical correction. The point is narrower: limited, noisy, and selectively observed evidence is not statistically entitled to induce unrestricted change. A constrained update can preserve capacities acquired from broader experience while further evidence accumulates.

The appropriate strength of regularization should decline as independent and robust evidence grows:

$$\lambda = \lambda(Q(E)), \quad \frac{d\lambda}{dQ} < 0,$$

where $Q(E)$ measures evidentiary quality. A system whose regularization never yields becomes dogmatic. A system with no regularization becomes reactive.

A.46 Feature drift and institutional drift

Let $\phi_t(x)$ be the representation assigned to input x at time t . Feature drift occurs when

$$D(\phi_{t+1}(x), \phi_t(x))$$

becomes large after exposure to a narrow sample. In institutions, the analogous representation may be a classification of behavior, personhood, threat, credibility, or permissible action.

An intensely publicized event can alter the institutional feature space itself. Behaviors formerly treated as ambiguous become coded as diagnostic; relationships acquire new presumptions;

neutral anomalies become threat indicators. Subsequent evidence is then interpreted through the altered representation.

This creates a feedback cycle:

$$E_t \rightarrow \Theta_{t+1} \rightarrow \phi_{t+1} \rightarrow \text{Select}(E_{t+1}) \rightarrow \Theta_{t+2}.$$

Early evidence does not merely influence a conclusion. It changes which later evidence the institution can recognize.

Procedural restraint limits this feedback by separating provisional classification from durable representational revision. Precedent, formal findings, and permanent records should generally require more support than temporary investigative hypotheses.

A.47 Posterior predictive checking

A model should be evaluated by asking whether data generated from its posterior predictive distribution resemble the observed data in relevant ways. The posterior predictive distribution is

$$P(\tilde{E} | E) = \int P(\tilde{E} | \theta) P(\theta | E) d\theta.$$

Choose a discrepancy statistic $T(E)$. The posterior predictive probability

$$P\left(T(\tilde{E}) \geq T(E) | E\right)$$

indicates whether the observed discrepancy is unusual under the fitted model.

In evidentiary reasoning, an explanation should be tested against features it was not constructed merely to restate. If a narrative claims coordinated planning, does it predict the temporal organization, communication structure, resource distribution, or error pattern that such planning would plausibly generate? If it claims independent spontaneous action, does the observed correlation exceed what that account predicts?

Posterior predictive failure should not automatically select a rival narrative. It may instead indicate that the entire model family is inadequate.

A.48 Calibration under adversarial conditions

Ordinary calibration assumes that the data-generating process does not strategically respond to the predictor. In adversarial environments, actors alter behavior after learning how they are evaluated.

Let a classifier use feature vector X and decision rule $d(X)$. An actor chooses transformation g to optimize

$$U_{\text{actor}}(g) = B(d(g(X))) - C(g),$$

where B is the benefit of obtaining a favorable decision and C the cost of changing observable features.

The deployed distribution becomes

$$P_{\text{deploy}}(X, Y) \neq P_{\text{predeploy}}(X, Y).$$

Surveillance and risk scoring can therefore produce an arms race in visibility. Less powerful actors may become legible in ways that expose them, while sophisticated actors learn to generate the surface features associated with compliance.

Institutional paranoia must be reflexive: it should suspect not only the subjects being evaluated but also the continued validity of the evaluation system.

A.49 Privacy and probabilistic inference

Privacy loss is not limited to the disclosure of recorded facts. Observed variables X may permit inference about unobserved property Z . The information that X carries about Z can be measured by mutual information:

$$I(X; Z) = \sum_{x,z} P(x, z) \log \frac{P(x, z)}{P(x)P(z)}.$$

If $I(X; Z) > 0$, observation of X reduces uncertainty about Z . Several individually weak signals may jointly become highly informative:

$$I(X_1, \dots, X_n; Z) \gg I(X_i; Z).$$

This is the probabilistic structure behind mosaic inference. Location traces, purchase histories, communication timing, social connections, and device metadata may jointly reveal medical, political, religious, occupational, or relational information never directly disclosed.

Privacy regulation limited to explicitly sensitive fields therefore misses inferential privacy. The actionable object is not only the stored datum but the posterior an institution can construct:

$$P(Z \mid X_{1:n}).$$

A person may protect every direct statement of Z while losing practical privacy because the surrounding traces determine Z with high probability.

A.50 Differential privacy and bounded disclosure

Differential privacy offers a formal guarantee limiting how much the inclusion of one person's data can change the distribution of released outputs. A randomized mechanism $\mathcal{M}$ is (ϵ, δ) -differentially private if, for neighboring datasets D and D' ,

$$P(\mathcal{M}(D) \in S) \leq e^\epsilon P(\mathcal{M}(D') \in S) + \delta$$

for all measurable output sets S .

The guarantee is relational. It limits the additional inferential exposure caused by an individual's participation. It does not guarantee that the output reveals nothing about that individual, especially when similar information is already inferable from the population or external data.

Differential privacy also introduces a privacy budget. Repeated releases compose, so apparently minor disclosures can accumulate. Under basic composition, mechanisms with parameters $\epsilon_1, \dots, \epsilon_k$ yield total privacy loss approximately bounded by

$$\varepsilon_{\text{total}} \leq \sum_{i=1}^k \varepsilon_i.$$

This formalizes an important principle of the book: privacy cannot be evaluated one disclosure at a time while ignoring combinatorial accumulation.

Differential privacy is not a general solution to legal transparency. It is most useful for aggregate statistical release. Courts, investigations, and public accountability often require particularized evidence. Its broader importance here lies in showing that publicity and privacy can sometimes be jointly designed rather than treated as absolute opposites.

A.51 Publicity as a change in the observation model

Publicity does not merely distribute evidence to more observers. It changes the process being observed. Once actors know they are visible, they may perform, conceal, signal, provoke, or curate behavior.

Let A be an actor's conduct and V the visibility regime. Then

$$P(A \mid V = 1) \neq P(A \mid V = 0).$$

A public recording is therefore not always a transparent sample of the behavior that would have occurred without recording. Publicity can deter misconduct, but it can also relocate misconduct outside the visible field or encourage symbolic actions designed for circulation.

The observer must reason about a reflexive loop:

$$V \rightarrow A \rightarrow R \rightarrow B \rightarrow V',$$

where visibility affects action, action produces records, records update public belief, and belief changes future visibility practices.

This complicates simple claims that more cameras reveal the underlying truth. Cameras alter the ecology whose truth is being investigated.

A.52 Public belief and common knowledge

A proposition can be privately known, publicly announced, or made common knowledge. If everyone knows H , that does not yet imply that everyone knows that everyone knows H . Common knowledge involves an indefinitely iterated structure:

$$K_1H, \quad K_2H, \quad K_1K_2H, \quad K_2K_1H, \dots$$

Publicity can transform behavior by creating higher-order beliefs even when it supplies little new first-order evidence. An accusation known privately by many people may have limited coordinating power. Once publicly announced, it becomes a reference point around which institutions, crowds, and adversaries can coordinate.

The public effect of information therefore cannot be measured solely by its contribution to $P(H \mid E)$. It also changes beliefs about what others believe and what actions they are likely to support.

A legally untested claim may consequently acquire institutional force through common knowledge before it acquires evidentiary warrant.

A.53 Belief, acceptance, and institutional commitment

An individual may assign high probability to a proposition without accepting it as a premise for action. Conversely, an institution may provisionally accept a proposition for a limited purpose without claiming that it is probably true in every sense.

Let $b(H) \in [0, 1]$ denote probabilistic belief and $A_c(H) \in \{0, 1\}$ denote acceptance in context c . Then

$$A_c(H) \neq \mathbf{1}[b(H) > \theta]$$

in general. Acceptance also depends upon the stakes, burden of proof, available alternatives, and procedural role.

A court may accept that sufficient evidence exists to authorize a search without accepting guilt. An employer may impose a temporary separation without making a final misconduct finding. A historian may treat a date as provisionally established while preserving a competing reconstruction.

Institutional pathology arises when acceptance migrates across contexts without preserving its original type:

$$A_{c_1}(H) \rightarrow A_{c_2}(H)$$

despite $c_1 \neq c_2$. A low-threshold investigative flag becomes a public assertion, then a reputational fact, then an input to later risk scoring. The probability has not necessarily changed, but the operational consequences have expanded.

A.54 The provenance of priors

Priors themselves have histories. Let the prior at time t be generated by earlier evidence $E_{<t}$, institutional transformations $T_{<t}$, and selection processes $S_{<t}$:

$$P_t(H) = \mathcal{G}(E_{<t}, T_{<t}, S_{<t}).$$

A prior may therefore contain accumulated knowledge, accumulated prejudice, or both. Calling it a prior does not settle its legitimacy.

A procedurally warranted prior should have recoverable provenance. One should be able to ask which observations produced it, whether those observations were representative, whether categories have changed, whether earlier errors were corrected, and whose experiences were excluded from the record.

The legal analogue of prior auditing is examination of precedent, institutional history, enforcement disparity, and category construction. A system that regularizes toward an inherited state without auditing that state may preserve exactly the feature drift of an earlier era.

Constrained updating must therefore operate at two timescales. Short-term regularization prevents a narrow new sample from causing impulsive change. Long-term review permits accumulated evidence to revise the structure toward which short-term updates are regularized.

A.55 A two-timescale model of institutional learning

Let θ_t denote fast operational beliefs and Φ_t denote slow institutional structure. New evidence updates θ_t rapidly:

$$\theta_{t+1} = \theta_t + \alpha_t g(E_t, \theta_t, \Phi_t),$$

where α_t is a comparatively large learning rate.

Institutional structure changes more slowly:

$$\Phi_{t+1} = \Phi_t + \beta_t h(E_{1:t}, \theta_{1:t}, \Phi_t),$$

with

$$0 < \beta_t \ll \alpha_t.$$

This arrangement permits provisional response without allowing each new event to rewrite the entire system. Structural change occurs after evidence accumulates across cases, undergoes review, and reveals persistent inadequacy.

The model also exposes two symmetric failures. If α_t is too large, operational judgment becomes reactive. If $\beta_t = 0$, structural injustice becomes permanent. Freedom requires both resistance to premature collapse and capacity for eventual transformation.

A.56 A formal distinction between updating and authorization

Let B_t be an institution's belief state and A_t its authorized action set. Evidence may update belief:

$$B_{t+1} = U(B_t, E_t).$$

Authorization is a separate map:

$$A_{t+1} = \Gamma(B_{t+1}, R, C),$$

where R is the applicable rights structure and C the procedural context.

It does not follow that every change in belief should change the action set:

$$B_{t+1} \neq B_t \not\Rightarrow A_{t+1} \neq A_t.$$

This separation is indispensable. Evidence can justify further investigation without justifying search, justify search without justifying charge, justify charge without justifying conviction, and justify conviction without justifying every possible punishment.

Probability informs belief. Law governs the transition from belief to authorized force.

A.57 A probabilistic formulation of premature collapse

Let $\mathcal{H}_t$ be the active set of hypotheses at time t , and let P_t be the current posterior distribution. Define a collapse operator

$$C_\tau(\mathcal{H}_\tau, P_\tau) = \hat{H}_\tau.$$

Collapse is premature when later admissible evidence substantially changes the warranted conclusion:

$$D(P_\tau, P_{\tau+\Delta}) > \epsilon$$

and the action authorized at τ was costly or irreversible.

A more prospective definition uses expected value of information. Collapse at time τ is premature when

$$EVS I_\tau > C_{\text{delay}, \tau}$$

and waiting remains institutionally possible.

This does not mean a decision is illegitimate whenever later evidence changes it. Institutions must act under uncertainty. Prematurity concerns a mismatch among evidentiary maturity, expected learning, action severity, and reversibility.

One can define a collapse-risk functional

$$\mathcal{C}(\tau) = P(D(P_\tau, P_{\tau+\Delta}) > \epsilon \mid E_{1:\tau}) \cdot I(a_\tau),$$

where $I(a_\tau)$ measures the irreversibility of the contemplated action. High collapse risk calls for stronger procedural restraint.

A.58 The probability of eventual recognition

Let a relevant feature F exist in an archive at time 0. It must be retained, retrieved, interpreted, validated, and escalated. Let the corresponding events be $R_1, \dots, R_5$. Then

$$P(\text{recognized by } t) = P(R_1 \cap \dots \cap R_5 \text{ by } t).$$

Under a simplifying conditional decomposition,

$$P(\text{recognized by } t) = P(R_1)P(R_2 \mid R_1)P(R_3 \mid R_1, R_2)P(R_4 \mid R_{1:3})P(R_5 \mid R_{1:4}).$$

Even moderately high probabilities compound. If each stage succeeds with probability 0.7, then the overall probability is

$$0.7^5 \approx 0.168.$$

Thus, a highly consequential feature may have less than a one-in-five chance of reaching institutional recognition despite no stage being exceptionally unreliable.

Over time, repeated search, technical improvement, new questions, or independent investigators may raise the cumulative probability. This produces long-tailed recognition times and explains why evidence recorded immediately can acquire significance ten or twenty years later.

A.59 Delayed evidence and retrospective updating

Suppose evidence E_t is recorded at event time t but recognized only at time $t + \Delta$. The later posterior is

$$P(H \mid E_{1:t+\Delta}),$$

but the historical question concerns what could reasonably have been believed or authorized at the earlier time. These are different quantities:

$$P(H \mid E_{\text{available at } t})$$

and

$$P(H \mid E_{\text{eventually recovered}}).$$

Retrospective certainty should not be projected backward as though all later evidence had been available initially. Conversely, the reasonableness of an earlier decision does not make its conclusion eternally true after new evidence emerges.

This yields two independent evaluations:

$$J_{\text{procedural}} = \text{Quality (decision given evidence then available)},$$

$$J_{\text{substantive}} = \text{Accuracy (decision given best evidence now available)}.$$

A decision can have been procedurally reasonable and substantively wrong. It can also have reached the right conclusion through an illegitimate procedure. These possibilities must remain distinct.

A.60 The asymmetry of accusation and correction

Let A be an accusation and C a later correction. Their public reach may satisfy

$$P(\text{exposed to } A) > P(\text{exposed to } C).$$

Among those exposed to both, anchoring may yield

$$P(H \mid A, C) \neq P(H \mid C, A).$$

A purely Bayesian ideal with fully specified evidence treats order as irrelevant because

$$P(H \mid A, C) = P(H \mid C, A).$$

Human and institutional cognition often violates this commutativity because the first item changes attention, interpretation, memory, and the hypothesis space through which the second is understood.

A realistic model must therefore allow the update operator to depend upon the current representation:

$$B_{t+1} = U(E_{t+1}; B_t, \Phi_t).$$

If the accusation changes Φ_t , the later correction is not processed by the same system that received the accusation. Correction may require representational repair, not merely addition of another evidence item.

A.61 Noncommutative updating

Classical conditioning on a fixed probability space is commutative with respect to the final evidence set:

$$P(H \mid E_1, E_2) = P(H \mid E_2, E_1).$$

Actual institutional updating may be noncommutative because evidence changes categories, search strategies, witness memories, resource allocations, and public incentives. Let U_E be an update operator. Then

$$U_{E_2}U_{E_1}(B_0) \neq U_{E_1}U_{E_2}(B_0).$$

This inequality should not be misunderstood as a rejection of probability theory. It indicates that the state being updated includes more than a fixed probability distribution. It includes the representation of hypotheses and the process by which subsequent evidence becomes observable.

Noncommutative updating provides the bridge from probability to the later geometric appendix. Different evidentiary paths can transport institutional judgment to different endpoints even when they contain nominally similar information.

A.62 Privacy, publicity, and asymmetric error

Privacy and publicity protect against different error distributions. Excessive secrecy raises the probability that institutional abuse remains undetected:

$$P(\text{undetected abuse} \mid \text{opacity}) \uparrow.$$

Excessive exposure raises the probability that incomplete personal information produces public or institutional misclassification:

$$P(\text{misclassification} \mid \text{unbounded exposure}) \uparrow.$$

The objective is not to minimize one risk without regard to the other. Let κ represent personal exposure and ρ institutional opacity. A general loss function is

$$L(\kappa, \rho) = L_{\text{concealed power}}(\rho) + L_{\text{exposed person}}(\kappa) + L_{\text{investigative failure}}(\kappa, \rho).$$

The interaction term matters. Some privacy can improve evidentiary quality by reducing coercion, performance, witness contamination, or retaliatory pressure. Some publicity can improve privacy by exposing unauthorized surveillance or forcing institutions to specify their information practices.

Privacy and publicity are therefore neither simple opposites nor universally substitutable goods.

A.63 Epistemic risk and moral risk

An epistemic risk is the possibility of holding a false or badly calibrated belief. A moral or political risk is the possibility of acting wrongly. The two are related but not identical.

Let e be epistemic error and a an action. The expected political harm is

$$E[L(a, H)] = \sum_H P(H | E) L(a, H).$$

A moderate epistemic error can produce enormous harm if the action is severe. A larger epistemic error may produce little harm if the belief remains private or supports only reversible inquiry.

This distinction explains why freedom cannot be protected merely by improving predictive accuracy. The institutional question is how error propagates into action:

$$\text{error} \rightarrow \text{classification} \rightarrow \text{authorization} \rightarrow \text{consequence}.$$

Procedural law introduces resistance along this chain. It does not presume that error can be eliminated at the point of observation.

A.64 The probabilistic meaning of disciplined paranoia

The power of paranoia is not captured by assigning high probability to malicious explanations. That would merely be a pessimistic prior. Disciplined paranoia instead operates at the level of model criticism.

It maintains four reservations:

$\mathcal{H}$ may be incomplete,

$P(E | H)$ may be misspecified,

E may have been selected non-randomly,

the action threshold may not follow from belief alone.

These reservations do not forbid conclusion. They require that confidence be indexed to its assumptions and that coercive consequences be indexed to the robustness of the conclusion.

A compact formal expression is

$$\text{Authorize}(a) \quad \text{only if} \quad \inf_{M \in \mathfrak{M}_{\text{adm}}} W(H, E, M) > \theta(a),$$

where W is a warrant function, $\mathfrak{M}_{\text{adm}}$ is a procedurally justified family of models, and $\theta(a)$ rises with the severity and irreversibility of a .

The infimum expresses caution across plausible interpretations. The restriction to admissible models prevents arbitrary possibilities from vetoing every action. The action-specific threshold prevents a single posterior confidence from licensing every consequence.

A.65 Summary propositions

The results of this appendix can be condensed into twelve propositions.

Proposition A.1: Representational incompleteness. A posterior distribution can be normalized over a formally exhaustive hypothesis set without that set being causally exhaustive.

Proposition A.2: Likelihood relativity. Evidence has no hypothesis-independent probative strength. Its force depends upon its relative probability under competing explanations.

Proposition A.3: Genealogical dependence. Repeated reports do not constitute repeated evidence when they descend from a common source or share a common error mechanism.

Proposition A.4: Selection dependence. Publicly available evidence is generally conditioned upon processes of recording, preservation, retrieval, ranking, and publication.

Proposition A.5: Precision under misspecification. Increasing data can produce increasing certainty around a systematically wrong model.

Proposition A.6: Robustness requirement. A conclusion is stronger when it survives plausible variations in priors, likelihoods, dependency structures, and selection models.

Proposition A.7: Temporal non-equivalence. The probability warranted by evidence available at the time of decision differs from the probability warranted by evidence recognized later.

Proposition A.8: Salience divergence. The probability that evidence receives attention need not increase with its legal or causal relevance.

Proposition A.9: Updating-authorization separation. Evidence may rationally change belief without authorizing a corresponding coercive action.

Proposition A.10: Severity-sensitive thresholds. The evidentiary threshold should ordinarily rise with the severity, duration, asymmetry, and irreversibility of the contemplated action.

Proposition A.11: Two-timescale learning. Institutions require fast provisional updating and slower structural revision; eliminating either produces reactivity or dogmatism.

Proposition A.12: Disciplined paranoia. Rational distrust concerns not only which hypothesis is true but whether the hypotheses, evidence-selection process, likelihoods, and action rules are themselves adequate.

A.66 Concluding interpretation

Probability theory clarifies why the abundance of records does not remove the need for legal or institutional restraint. A posterior is not produced by evidence alone. It is produced by evidence situated within a hypothesis space, a likelihood model, a provenance structure, a selection process, a reference class, and a rule governing when inquiry stops. Each element can fail while the final calculation remains internally correct.

This does not make probabilistic reasoning futile. It makes its conditions visible. Properly used, probability separates relative support from mere consistency, exposes double-counted sources, represents delayed evidence, measures sensitivity to assumptions, and distinguishes the degree of belief from the authority to act. Improperly used, it conceals structural uncertainty beneath numerical precision.

The legal requirement that claims pass through evidence, contestation, and standards of proof is therefore not an antiquated substitute for mathematical rationality. It is a social response to uncertainties that no single probability assignment can conclusively absorb. Procedure allows assumptions to be challenged by persons who were not represented in the original model. It

permits new hypotheses to enter, provenance to be disputed, dependencies to be revealed, and action to be withheld when confidence is fragile.

The power of paranoia lies in remembering that even a highly probable account remains an account. Its discipline lies in specifying what would warrant belief, what would warrant revision, and what degree of belief would warrant which action.

Appendix B

Causal Inference and Counterfactual Structure

Appendix C

Information Theory, Search, and Compression

Appendix D

Dynamical Systems, Geometry, and Holonomy

Appendix E

Decision Theory and Institutional Loss

Conclusion